

$$\begin{aligned} \mathbf{R}_{12} &= 5 \text{ in } @ \angle 210^\circ; & R_{12_x} &= -4.33, & R_{12_y} &= -2.50 \\ \mathbf{R}_P &= 5 \text{ in } @ \angle 30^\circ; & R_{P_x} &= +4.33, & R_{P_y} &= +2.50 \end{aligned} \quad (b)$$

- 4 Calculate the x and y components of the acceleration of the CG in this coordinate system:

$$\mathbf{a}_G = 2001 @ \angle 208^\circ; \quad a_{G_x} = -1766.78, \quad a_{G_y} = -939.41 \quad (c)$$

- 5 Calculate the x and y components of the external force at P in this coordinate system:

$$\mathbf{F}_P = 40 @ \angle 0^\circ; \quad F_{P_x} = 40, \quad F_{P_y} = 0 \quad (d)$$

- 6 Substitute these given and calculated values into the matrix equation 11.4

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2.50 & -4.33 & 1 \end{bmatrix} \times \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} (0.01)(-1766.78) - 40 \\ (0.01)(-939.41) - 0 \\ (0.08)(15) - \{(4.33)(0) - (2.5)(40)\} \end{bmatrix} \quad (e)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2.50 & -4.33 & 1 \end{bmatrix} \times \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} -57.67 \\ -9.39 \\ 101.2 \end{bmatrix}$$

- 7 Solve this system either by inverting matrix $\mathbf{A}$ and premultiplying that inverse times matrix $\mathbf{C}$ using a pocket calculator such as the HP-15c or by inputting the values for matrices $\mathbf{A}$ and $\mathbf{C}$ to program MATRIX provided with this text.

Program MATRIX gives the following solution:

$$F_{12_x} = -57.67 \text{ lb}, \quad F_{12_y} = -9.39 \text{ lb}, \quad T_{12} = 204.72 \text{ lb-in} \quad (f)$$

Converting the force to polar coordinates:

$$\mathbf{F}_{12} = 58.43 @ \angle 189.25^\circ \quad (g)$$

Read the disk file E011-01.mat into program MATRIX to exercise this example.

11.3 FORCE ANALYSIS OF A THREEBAR CRANK-SLIDE LINKAGE

When there is more than one link in the assembly, the solution simply requires that the three equations 11.1b be written for each link and then solved simultaneously. Figure 11-2a shows a threebar crank-slide linkage. This linkage has been simplified from the fourbar slider-crank (see Figure 11-4) by replacing the kinematically redundant slider block (link 4) with a half joint as shown. This linkage transformation reduces the number of links to three with no change in degree of freedom (see Section 2.9, p. 40). Only links 2 and 3 are moving. Link 1 is ground. Thus we should expect to have six equations in six unknowns (three per moving link).



Dynamic force analysis of a slider-crank linkage

The *CG* of each link is initially defined by a position vector rooted at one pin joint whose angle is measured with respect to the line of centers of the link in the local, rotating coordinate system (LRCS) x' , y' . This is the most convenient way to establish the *CG* location since the link line of centers is the kinematic definition of the link. However, we will need to define the link's dynamic parameters and force locations with respect to a local, nonrotating coordinate system (LNCS) x , y located at its *CG* and which is always parallel to the global coordinate system (GCS) XY . The position vector locations of all attachment points of other links and points of application of external forces must

be defined with respect to the link's LNCS. Note that these kinematic and applied force data must be available for all positions of the linkage for which a force analysis is desired. In the following discussion and examples, only one linkage position will be addressed. The process is identical for each succeeding position and only the calculations must be repeated. Obviously, a computer will be a valuable aid in accomplishing the task.

Link 2 in Figure 11-2b shows forces acting on it at each pin joint, designated F_{12} and F_{32} . By convention their subscripts denote the force that the adjoining link is exerting *on* the link being analyzed; that is, F_{12} is the force of 1 *on* 2 and F_{32} is the force of 3 *on* 2. Obviously there is also an equal and opposite force at each of these pins which would be designated as F_{21} and F_{23} , respectively. The choice of which of the members of these pairs of forces to be solved for is arbitrary. As long as proper bookkeeping is done, their identities will be maintained.

When we move to link 3, we maintain the same convention of showing forces acting *on* the link in its free-body diagram. Thus at instant center I_{23} we show F_{23} acting on link 3. However, because we showed force F_{32} acting at the same point on link 2, this introduces an additional unknown to the problem for which we need an additional equation. The equation is available from Newton's third law:

$$\mathbf{F}_{23} = -\mathbf{F}_{32} \quad (11.5)$$

Thus we are free to substitute the negative reaction force for any action force at any joint. This has been done on link 3 in the figure in order to reduce the unknown forces at that joint to one, namely F_{32} . The same procedure is followed at each joint with one of the action-reaction forces arbitrarily chosen to be solved for and its negative reaction applied to the mating link.

The naming convention used for the position vectors (R_{ap}) which locate the pin joints with respect to the CG in the link's nonrotating local coordinate system is as follows. The first subscript (a) denotes the adjoining link to which the position vector points. The second subscript (p) denotes the parent link to which the position vector belongs. Thus in the case of link 2 in Figure 11-2b, vector R_{12} locates the attachment point of link 1 to link 2, and R_{32} the attachment point of link 3 to link 2. Note that in some cases these subscripts will match those of the pin forces shown acting at those points, but where the negative reaction force has been substituted as described above, the subscript order of the force and its position vector will not agree. This can lead to confusion and must be carefully watched for typographical errors when setting up the problem.

Any external forces acting on the links are located in similar fashion with a position vector to a point on the line of application of the force. This point is given the same letter subscript as that of the external force. Link 3 in the figure shows such an external force F_p acting on it at point P . The position vector R_p locates that point with respect to the CG. It is important to note that the CG of each link is consistently taken as the point of reference for all forces acting on that link. Left to its own devices, an unconstrained body in complex motion will spin about its own CG; thus we analyze its linear acceleration at that point and apply the angular acceleration about the CG as a center.

Equations 11.1 are now written for each moving link. For link 2, with the cross products expanded:

$$\begin{aligned}
F_{12_x} + F_{32_x} &= m_2 a_{G_2_x} \\
F_{12_y} + F_{32_y} &= m_2 a_{G_2_y} \\
T_{12} + (R_{12_x} F_{12_y} - R_{12_y} F_{12_x}) + (R_{32_x} F_{32_y} - R_{32_y} F_{32_x}) &= I_{G_2} \alpha_2
\end{aligned} \tag{11.6a}$$

For link 3, with the cross products expanded, note the substitution of the reaction force $-\mathbf{F}_{32}$ for $\mathbf{F}_{23}$:

$$\begin{aligned}
F_{13_x} - F_{32_x} + F_{P_x} &= m_3 a_{G_3_x} \\
F_{13_y} - F_{32_y} + F_{P_y} &= m_3 a_{G_3_y} \\
(R_{13_x} F_{13_y} - R_{13_y} F_{13_x}) - (R_{23_x} F_{32_y} - R_{23_y} F_{32_x}) + (R_{P_x} F_{P_y} - R_{P_y} F_{P_x}) &= I_{G_3} \alpha_3
\end{aligned} \tag{11.6b}$$

Note also that T_{12} , the source torque, only appears in the equation for link 2 as that is the driver crank to which the motor is attached. Link 3 has no externally applied torque but does have an external force $\mathbf{F}_P$ which might be due to whatever link 3 is pushing on to do its external work.

There are seven unknowns present in these six equations, F_{12_x} , F_{12_y} , F_{32_x} , F_{32_y} , F_{13_x} , F_{13_y} , and T_{12} . But, F_{13_y} is due only to friction at the joint between link 3 and link 1. We can write a relation for the friction force f at that interface such as $f = \pm \mu N$, where $\pm \mu$ is a known coefficient of coulomb friction. The friction force always opposes motion. The kinematic analysis will provide the velocity of the link at the sliding joint. The direction of f will always be the opposite of this velocity. Note that μ is a nonlinear function which has a discontinuity at zero velocity; thus at the linkage positions where velocity is zero, the inclusion of μ in these linear equations is not valid. (See Figure 10-5a, p. 502.) In this example, the normal force N is equal to F_{13_x} and the friction force f is equal to F_{13_y} . For linkage positions with nonzero velocity, we can eliminate F_{13_y} by substituting into equation 11.6b,

$$F_{13_y} = \mu F_{13_x} \tag{11.6c}$$

where the sign of F_{13_y} is taken as the opposite of the sign of the velocity at that point. We are then left with six unknowns in equations 11.6 and can solve them simultaneously. We also rearrange equations 11.6a and 11.6b to put all known terms on the right side.

$$\begin{aligned}
F_{12_x} + F_{32_x} &= m_2 a_{G_2_x} \\
F_{12_y} + F_{32_y} &= m_2 a_{G_2_y} \\
T_{12} + R_{12_x} F_{12_y} - R_{12_y} F_{12_x} + R_{32_x} F_{32_y} - R_{32_y} F_{32_x} &= I_{G_2} \alpha_2
\end{aligned} \tag{11.6d}$$

$$\begin{aligned}
F_{13_x} - F_{32_x} &= m_3 a_{G_3_x} - F_{P_x} \\
\pm \mu F_{13_x} - F_{32_y} &= m_3 a_{G_3_y} - F_{P_y} \\
(\pm \mu R_{13_x} - R_{13_y}) F_{13_x} - R_{23_x} F_{32_y} + R_{23_y} F_{32_x} &= I_{G_3} \alpha_3 - R_{P_x} F_{P_y} + R_{P_y} F_{P_x}
\end{aligned}$$

Putting these six equations in matrix form we get:

$$\begin{bmatrix}
 1 & 0 & 1 & 0 & 0 & 0 \\
 0 & 1 & 0 & 1 & 0 & 0 \\
 -R_{12y} & R_{12x} & -R_{32y} & R_{32x} & 0 & 1 \\
 0 & 0 & -1 & 0 & 1 & 0 \\
 0 & 0 & 0 & -1 & \mu & 0 \\
 0 & 0 & R_{23y} & -R_{23x} & (\mu R_{13x} - R_{13y}) & 0
 \end{bmatrix} \times \begin{bmatrix} F_{12x} \\ F_{12y} \\ F_{32x} \\ F_{32y} \\ F_{13x} \\ T_{12} \end{bmatrix} =$$

(11.7)

$$\begin{bmatrix}
 m_2 a_{G_2x} \\
 m_2 a_{G_2y} \\
 I_{G_2} \alpha_2 \\
 m_3 a_{G_3x} - F_{Px} \\
 m_3 a_{G_3y} - F_{Py} \\
 I_{G_3} \alpha_3 - R_{Px} F_{Py} + R_{Py} F_{Px}
 \end{bmatrix}$$

This system can be solved by using program MATRIX or any other matrix solving calculator. As an example of this solution consider the following linkage data.

EXAMPLE 11-2

Dynamic Force Analysis of a Threebar Crank-Slide Linkage with Half Joint. (See Figure 11-2, p. 526.)

Given: The 5-in long crank (link 2) shown weighs 2 lb. Its CG is at 3 in and 30° from the line of centers. Its mass moment of inertia about its CG is 0.05 lb-in-sec². Its acceleration is defined in its LNCS, x, y . Its kinematic data are:

θ_2 deg	ω_2 rad/sec	α_2 rad/sec ²	a_{G_2} in/sec ²
60	30	-10	2700.17 @ -89.4°

The coupler (link 3) is 15 in long and weighs 4 lb. Its CG is at 9 in and 45° from the line of centers. Its mass moment of inertia about its CG is 0.10 lb-in-sec². Its acceleration is defined in its LNCS, x, y . Its kinematic data are:

θ_3 deg	ω_3 rad/sec	α_3 rad/sec ²	a_{G_3} in/sec ²
99.59	-8.78	-136.16	3453.35 @ 254.4°

The sliding joint on link 3 has a velocity of 96.95 in/sec in the +Y direction.

There is an external force of 50 lb at -45°, applied at point P which is located at 2.7 in and 101° from the CG of link 3, measured in the link's embedded, rotating coordinate system or LRCS x', y' (origin at A and x axis from A to B). The coefficient of friction μ is 0.2.

Find: The forces $\mathbf{F}_{12}$, $\mathbf{F}_{32}$, $\mathbf{F}_{13}$ at the joints and the driving torque T_{12} needed to maintain motion with the given acceleration for this instantaneous position of the link.

Solution:

- 1 Convert the given weights to proper mass units, in this case blobs:

$$mass_{link2} = \frac{weight}{g} = \frac{2 \text{ lb}}{386 \text{ in/sec}^2} = 0.0052 \text{ blobs} \quad (a)$$

$$mass_{link3} = \frac{weight}{g} = \frac{4 \text{ lb}}{386 \text{ in/sec}^2} = 0.0104 \text{ blobs} \quad (b)$$

- 2 Set up a local, nonrotating xy coordinate system (LNCS) at the CG of each link, and draw all applicable position and force vectors acting within or on that system as shown in Figure 11-2. Draw a free-body diagram of each moving link as shown.
- 3 Calculate the x and y components of the position vectors $\mathbf{R}_{12}$, $\mathbf{R}_{32}$, $\mathbf{R}_{23}$, $\mathbf{R}_{13}$, and $\mathbf{R}_P$ in the LNCS coordinate system:

$$\begin{array}{lll} \mathbf{R}_{12} = 3.00 @ \angle 270.0^\circ; & R_{12_x} = 0, & R_{12_y} = -3.0 \\ \mathbf{R}_{32} = 2.83 @ \angle 28.0^\circ; & R_{32_x} = 2.500, & R_{32_y} = 1.333 \\ \mathbf{R}_{23} = 9.00 @ \angle 324.5^\circ; & R_{23_x} = 7.329, & R_{23_y} = -5.224 \\ \mathbf{R}_{13} = 10.72 @ \angle 63.14^\circ; & R_{13_x} = 4.843, & R_{13_y} = 9.563 \\ \mathbf{R}_P = 2.70 @ \angle 201.0^\circ; & R_{P_x} = -2.521, & R_{P_y} = -0.968 \end{array} \quad (c)$$

These position vector angles are measured with respect to the LNCS which is always parallel to the global coordinate system (GCS), making the angles the same in both systems.

- 4 Calculate the x and y components of the acceleration of the CG s of all moving links in the global coordinate system:

$$\begin{array}{lll} \mathbf{a}_{G_2} = 2700.17 @ \angle -89.4^\circ; & a_{G_2_x} = 28.28, & a_{G_2_y} = -2700 \\ \mathbf{a}_{G_3} = 3453.35 @ \angle 254.4^\circ; & a_{G_3_x} = -930.82, & a_{G_3_y} = -3325.54 \end{array} \quad (d)$$

- 5 Calculate the x and y components of the external force at P in the global coordinate system:

$$\mathbf{F}_P = 50 @ \angle -45^\circ; \quad F_{P_x} = 35.36, \quad F_{P_y} = -35.36 \quad (e)$$

- 6 Substitute these given and calculated values into the matrix equation 11.7.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 3 & 0 & -1.333 & 2.5 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0.2 & 0 \\ 0 & 0 & -5.224 & -7.329 & [(0.2)4.843 - (9.563)] & 0 \end{bmatrix} \times \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{13_x} \\ T_{12} \end{bmatrix} = \begin{bmatrix} (0.005)(28.28) \\ (0.005)(-2700) \\ (0.05)(-10) \\ (0.01)(-930.82) - 35.36 \\ (0.01)(-3325.54) - (-35.36) \\ (0.1)(-136.16) - (-2.521)(-35.36) + (-0.968)(35.36) \end{bmatrix} = \begin{bmatrix} 0.141 \\ -13.500 \\ -.500 \\ -44.668 \\ 2.105 \\ -136.987 \end{bmatrix} \quad (f)$$

- 7 Solve this system either by inverting matrix **A** and premultiplying that inverse times matrix **C** using a pocket calculator such as the HP-15c, or by inputting the values for matrices **A** and **C** to program MATRIX provided with this text which gives the following solution:

$$\begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{13_x} \\ T_{12} \end{bmatrix} = \begin{bmatrix} -39.232 \\ -10.336 \\ 39.373 \\ -3.164 \\ -5.295 \\ 177.590 \end{bmatrix} \quad (g)$$

Converting the forces to polar coordinates:

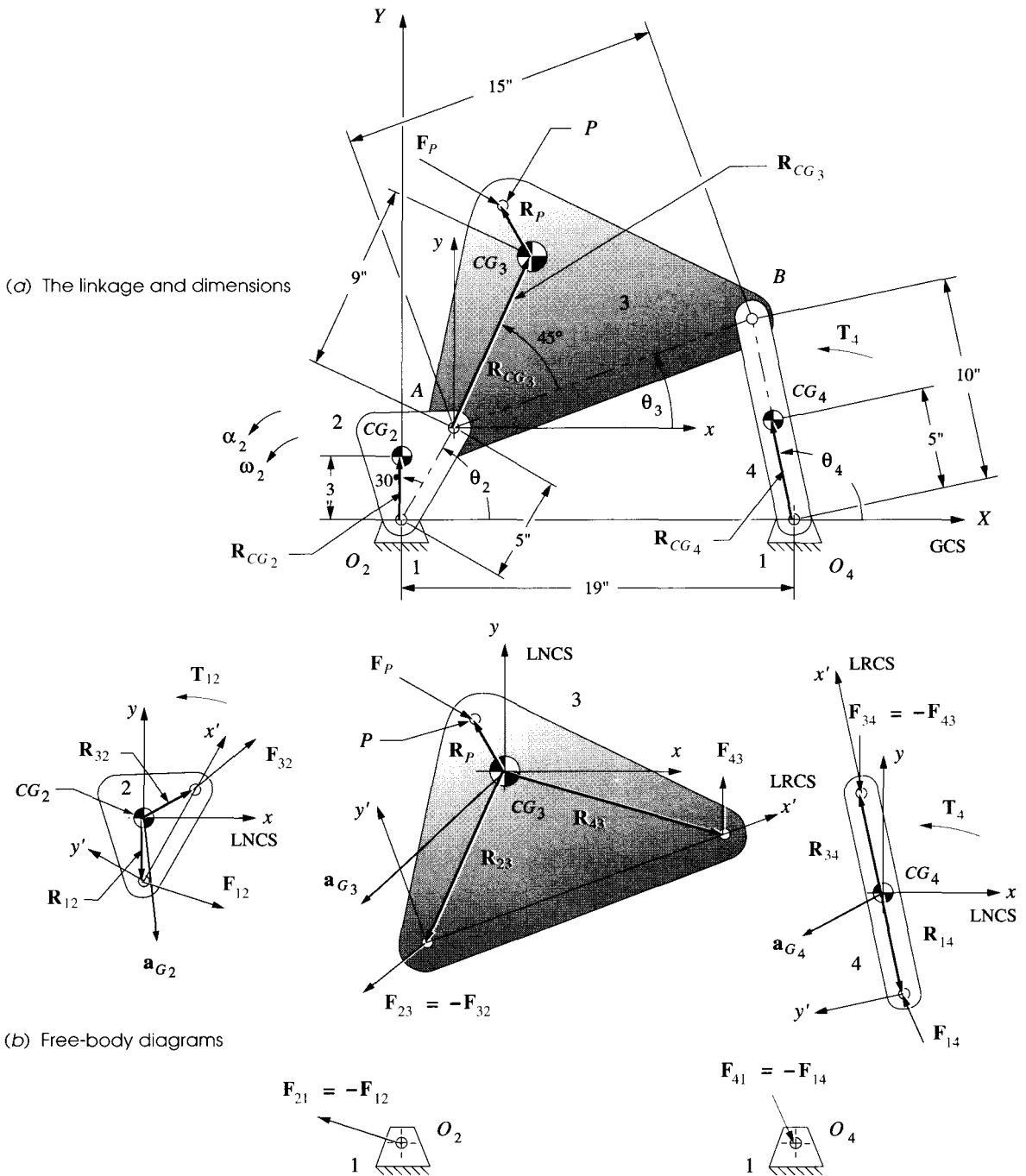
$$\begin{aligned} \mathbf{F}_{12} &= 40.57 \text{ lb @ } \angle 194.76^\circ \\ \mathbf{F}_{32} &= 39.50 \text{ lb @ } \angle -4.60^\circ \\ \mathbf{F}_{13} &= 5.40 \text{ lb @ } \angle 191.31^\circ \end{aligned} \quad (h)$$

Read the disk file E011-02.mat into program MATRIX to exercise this example.

11.4 FORCE ANALYSIS OF A FOURBAR LINKAGE

Figure 11-3a shows a fourbar linkage. All dimensions of link lengths, link positions, locations of the links' CGs, linear accelerations of those CGs, and link angular accelerations and velocities have been previously determined from a kinematic analysis. We now wish to find the forces acting at all the pin joints of the linkage for one or more positions. The procedure is exactly the same as that used in the above two examples. This linkage has three moving links. Equation 11.1 provides three equations for any link or rigid body in motion. We should expect to have nine equations in nine unknowns for this problem.

Figure 11-3b shows the free-body diagrams for all links, with all forces shown. Note that an external force $\mathbf{F}_P$ is shown acting on link 3 at point *P*. Also an external torque T_4 is shown acting on link 4. These external loads are due to some other mechanism (de-

**FIGURE 11-3**

Dynamic force analysis of a fourbar linkage. (See also Figure P11-2, p. 561)

vice, person, thing, etc.) pushing or twisting against the motion of the linkage. Any link can have any number of external loads and torques acting on it. Only one external torque and one external force are shown here to serve as examples of how they are handled in the computation. (Note that a more complicated force system, if present, could also be reduced to the combination of a single force and torque on each link.)

To solve for the pin forces it is necessary that these applied external forces and torques be defined for all positions of interest. We will solve for one member of the pair of action-reaction forces at each joint, and also for the driving torque T_{12} needed to be supplied at link 2 in order to maintain the kinematic state as defined. The force subscript convention is the same as that defined in the previous example. For example, F_{12} is the force of 1 on 2 and F_{32} is the force of 3 on 2. The equal and opposite forces at each of these pins are designated F_{21} and F_{23} , respectively. All the unknown forces in the figure are shown at arbitrary angles and lengths as their true values are still to be determined.

The linkage kinematic parameters are defined with respect to a global XY system (GCS) whose origin is at the driver pivot O_2 and whose X axis goes through link 4's fixed pivot O_4 . The mass (m) of each link, the location of its CG , and its mass moment of inertia (I_G) about that CG are also needed. The CG of each link is initially defined within each link with respect to a local moving and rotating axis system (LRCS) embedded in the link because the CG is an unchanging physical property of the link. The origin of this x',y' axis system is at one pin joint and the x' axis is the line of centers of the link. The CG position within the link is defined by a position vector in this LRCS. The instantaneous location of the CG can easily be determined for each dynamic link position by adding the angle of the internal CG position vector to the current GCS angle of the link.

We need to define each link's dynamic parameters and force locations with respect to a local, moving, but nonrotating axis system (LNCS) x,y located at its CG as shown on each free-body diagram in Figure 11-3b. The position vector locations of all attachment points of other links and points of application of external forces must be defined with respect to this LNCS axis system. These kinematic and applied force data differ for each position of the linkage. In the following discussion and examples, only one linkage position will be addressed. The process is identical for each succeeding position.

Equations 11.1 (p. 522) are now written for each moving link. For link 2, the result is identical to that done for the slider-crank example in equation 11.6a (p. 528).

$$\begin{aligned} F_{12_x} + F_{32_x} &= m_2 a_{G_2_x} \\ F_{12_y} + F_{32_y} &= m_2 a_{G_2_y} \\ T_{12} + (R_{12_x} F_{12_y} - R_{12_y} F_{12_x}) + (R_{32_x} F_{32_y} - R_{32_y} F_{32_x}) &= I_{G_2} \alpha_2 \end{aligned} \quad (11.8a)$$

For link 3, with substitution of the reaction force $-F_{32}$ for F_{23} , the result is similar to equation 11.6b with some subscript changes to reflect the presence of link 4.

$$\begin{aligned} F_{43_x} - F_{32_x} + F_{P_x} &= m_3 a_{G_3_x} \\ F_{43_y} - F_{32_y} + F_{P_y} &= m_3 a_{G_3_y} \\ (R_{43_x} F_{43_y} - R_{43_y} F_{43_x}) - (R_{23_x} F_{32_y} - R_{23_y} F_{32_x}) + (R_{P_x} F_{P_y} - R_{P_y} F_{P_x}) &= I_{G_3} \alpha_3 \end{aligned} \quad (11.8b)$$

For link 4, substituting the reaction force $-F_{43}$ for F_{34} , a similar set of equations 11.1 can be written:

$$\begin{aligned} F_{14_x} - F_{43_x} &= m_4 a_{G_{4_x}} \\ F_{14_y} - F_{43_y} &= m_4 a_{G_{4_y}} \\ (R_{14_x} F_{14_y} - R_{14_y} F_{14_x}) - (R_{34_x} F_{43_y} - R_{34_y} F_{43_x}) + T_4 &= I_{G_4} \alpha_4 \end{aligned} \quad (11.8c)$$

Note again that T_{12} , the source torque, only appears in the equation for link 2 as that is the driver crank to which the motor is attached. Link 3, in this example, has no externally applied torque (though it could have) but does have an external force F_P . Link 4, in this example, has no external force acting on it (though it could have) but does have an external torque T_4 . (The driving link 2 could also have an externally applied force on it though it usually does not.) There are nine unknowns present in these nine equations, F_{12_x} , F_{12_y} , F_{32_x} , F_{32_y} , F_{43_x} , F_{43_y} , F_{14_x} , F_{14_y} , and T_{12} , so we can solve them simultaneously. We rearrange terms in equations 11.8 to put all known constant terms on the right side and then put them in matrix form.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -R_{12_y} & R_{12_x} & -R_{32_y} & R_{32_x} & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & R_{23_y} & -R_{23_x} & -R_{43_y} & R_{43_x} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & R_{34_y} & -R_{34_x} & -R_{14_y} & R_{14_x} & 0 \end{bmatrix} \times \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_x} \\ F_{14_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G_{2_x}} \\ m_2 a_{G_{2_y}} \\ I_{G_2} \alpha_2 \\ m_3 a_{G_{3_x}} - F_{P_x} \\ m_3 a_{G_{3_y}} - F_{P_y} \\ I_{G_3} \alpha_3 - R_{P_x} F_{P_y} + R_{P_y} F_{P_x} \\ m_4 a_{G_{4_x}} \\ m_4 a_{G_{4_y}} \\ I_{G_4} \alpha_4 - T_4 \end{bmatrix} \quad (11.9)$$

This system can be solved by using program MATRIX or any matrix solving calculator. As an example of this solution consider the following linkage data.

EXAMPLE 11-3

Dynamic Force Analysis of a Fourbar Linkage. (See Figure 11-3, p. 532)

Given: The 5-in-long crank (link 2) shown weighs 1.5 lb. Its CG is at 3 in at $+30^\circ$ from the line of centers. Its mass moment of inertia about its CG is 0.4 lb-in-sec². Its kinematic data are:

θ_2 deg	ω_2 rad/sec	α_2 rad/sec ²	a_{G_2} in/sec ²
60	25	-40	1878.84 @ 273.66°

The coupler (link 3) is 15 in long and weighs 7.7 lb. Its CG is at 9 in at 45° off the line of centers. Its mass moment of inertia about its CG is 1.5 lb-in-sec². Its kinematic data are:

θ_3 deg	ω_3 rad/sec	α_3 rad/sec ²	a_{G_3} in/sec ²
20.92	-5.87	120.9	3646.1 @ 226.5°

There is an external force of 80 lb at 330° on link 3, applied at point *P* which is located 3 in at 100° from the CG of link 3. There is an external torque on link 4 of 120 lb-in. The ground link is 19 in long. The rocker (link 4) is 10 in long and weighs 5.8 lb. Its CG is at 5 in at 0° off the line of centers. Its mass moment of inertia about its CG is 0.8 lb-in-sec². Its kinematic data are:

θ_4 deg	ω_4 rad/sec	α_4 rad/sec ²	a_{G_4} in/sec ²
104.41	7.93	276.29	1416.8 @ 207.2°

Find: The forces $\mathbf{F}_{12}$, $\mathbf{F}_{32}$, $\mathbf{F}_{43}$, $\mathbf{F}_{14}$, at the joints and the driving torque $\mathbf{T}_{12}$ needed to maintain motion with the given acceleration for this instantaneous position of the link.

Solution:

- 1 Convert the given weight to proper mass units, in this case blobs:

$$mass_{link2} = \frac{weight}{g} = \frac{1.5 \text{ lb}}{386 \text{ in/sec}^2} = 0.004 \text{ blobs} \quad (a)$$

$$mass_{link3} = \frac{weight}{g} = \frac{7.7 \text{ lb}}{386 \text{ in/sec}^2} = 0.020 \text{ blobs} \quad (b)$$

$$mass_{link4} = \frac{weight}{g} = \frac{5.8 \text{ lb}}{386 \text{ in/sec}^2} = 0.015 \text{ blobs} \quad (c)$$

- 2 Set up an LNCS *xy* coordinate system at the CG of each link, and draw all applicable vectors acting on that system as shown in the figure. Draw a free-body diagram of each moving link as shown.
- 3 Calculate the *x* and *y* components of the position vectors $\mathbf{R}_{12}$, $\mathbf{R}_{32}$, $\mathbf{R}_{23}$, $\mathbf{R}_{43}$, $\mathbf{R}_{34}$, $\mathbf{R}_{14}$, and $\mathbf{R}_P$ in the link's LNCS. $\mathbf{R}_{43}$, $\mathbf{R}_{34}$, and $\mathbf{R}_{14}$ will have to be calculated from the given link geometry data using the law of cosines and law of sines. Note that the current value of link 3's position angle (θ_3) in the GCS must be added to the angles of all position vectors before creating their *x,y* components in the LNCS if their angles were originally measured with respect to the link's embedded, local rotating coordinate system (LRCS).

$$\begin{aligned}
\mathbf{R}_{12} &= 3.00 @ \angle 270.00^\circ; & R_{12_x} &= 0.000, & R_{12_y} &= -3 \\
\mathbf{R}_{32} &= 2.83 @ \angle 28.00^\circ; & R_{32_x} &= 2.500, & R_{32_y} &= 1.333 \\
\mathbf{R}_{23} &= 9.00 @ \angle 245.92^\circ; & R_{23_x} &= -3.672, & R_{23_y} &= -8.217 \\
\mathbf{R}_{43} &= 10.72 @ \angle -15.46^\circ; & R_{43_x} &= 10.332, & R_{43_y} &= -2.858 \\
\mathbf{R}_{34} &= 5.00 @ \angle 104.41^\circ; & R_{34_x} &= -1.244, & R_{34_y} &= 4.843 \\
\mathbf{R}_{14} &= 5.00 @ \angle 284.41^\circ; & R_{14_x} &= 1.244, & R_{14_y} &= -4.843 \\
\mathbf{R}_P &= 3.00 @ \angle 120.92^\circ; & R_{P_x} &= -1.542, & R_{P_y} &= 2.574
\end{aligned} \quad (d)$$

- 4 Calculate the x and y components of the acceleration of the CGs of all moving links in the global coordinate system (GCS):

$$\begin{aligned}
\mathbf{a}_{G_2} &= 1878.84 @ \angle 273.66^\circ; & a_{G_{2_x}} &= 119.94, & a_{G_{2_y}} &= -1875.01 \\
\mathbf{a}_{G_3} &= 3646.10 @ \angle 226.51^\circ; & a_{G_{3_x}} &= -2509.35, & a_{G_{3_y}} &= -2645.23 \\
\mathbf{a}_{G_4} &= 1416.80 @ \angle 207.24^\circ; & a_{G_{4_x}} &= -1259.67, & a_{G_{4_y}} &= -648.50
\end{aligned} \quad (e)$$

- 5 Calculate the x and y components of the external force at P in the GCS:

$$\mathbf{F}_{P3} = 80 @ \angle 330^\circ; \quad F_{P3_x} = 69.28, \quad F_{P3_y} = -40.00 \quad (f)$$

- 6 Substitute these given and calculated values into the matrix equation 11.9 (p. 534).

$$\begin{bmatrix}
1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
3 & 0 & -1.330 & 2.5 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & -8.217 & 3.673 & 2.861 & 10.339 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 4.843 & 1.244 & 4.843 & 1.244 & 0
\end{bmatrix} \times \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_x} \\ F_{14_y} \\ T_{12} \end{bmatrix} =$$

(g)

$$\begin{bmatrix}
(0.004)(119.94) \\
(0.004)(-1875.01) \\
(0.4)(-40) \\
(0.02)(-2509.35) - (69.28) \\
(0.02)(-2645.23) - (-40) \\
(1.5)(120.9) - [(-1.542)(-40) - (2.574)(69.28)] \\
(0.015)(-1259.67) \\
(0.015)(-648.50) \\
(0.8)(276.29) - (120)
\end{bmatrix} = \begin{bmatrix} 0.480 \\ -7.500 \\ -16.000 \\ -119.465 \\ -12.908 \\ 298.003 \\ -18.896 \\ -9.727 \\ 101.031 \end{bmatrix}$$

- 7 Solve this system either by inverting matrix A and premultiplying that inverse times matrix C using a pocket calculator such as the HP-28, or by inputting the values for matrices A and C to program MATRIX provided with this text which gives the following solution:

$$\begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_x} \\ F_{14_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} -117.65 \\ -107.84 \\ 118.13 \\ 100.34 \\ -1.34 \\ 87.43 \\ -20.23 \\ 77.71 \\ 243.23 \end{bmatrix} \quad (h)$$

Converting the forces to polar coordinates:

$$\begin{aligned} \mathbf{F}_{12} &= 159.60 \text{ lb @ } \angle 222.52^\circ \\ \mathbf{F}_{32} &= 154.99 \text{ lb @ } \angle 40.35^\circ \\ \mathbf{F}_{43} &= 87.44 \text{ lb @ } \angle 90.88^\circ \\ \mathbf{F}_{14} &= 80.30 \text{ lb @ } \angle 104.59^\circ \end{aligned} \quad (i)$$

- 8 The pin-force magnitudes in (i) are needed to size the pivot pins and links against failure and to select pivot bearings that will last for the required life of the assembly. The driving torque T_{12} defined in (h) is needed to select a motor or other device capable of supplying the power to drive the system. See Section 2.16 (p. 60) for a brief discussion of motor selection. Issues of stress calculation and failure prevention are beyond the scope of this text, but note that those calculations cannot be done until a good estimate of the dynamic forces and torques on the system has been made by methods such as those shown in this example.

This solves the linkage for one position. A new set of values can be put into the A and C matrices for each position of interest at which a force analysis is needed. Read the disk file EII-03.mat into program MATRIX to exercise this example. The disk file EII-03.4br can also be read into program FOURBAR which will run the linkage through a series of positions starting with the stated parameters as initial conditions. The linkage will slow to a stop and then run in reverse due to the negative acceleration. The matrix of equation (g) can be seen within FOURBAR using *Dynamics/Solve/Show Matrix*.

It is worth noting some general observations about this method at this point. The solution is done using cartesian coordinates of all forces and position vectors. Before being placed in the matrices, these vector components must be defined in the global coordinate system (GCS) or in nonrotating, local coordinate systems, parallel to the global coordinate system, with their origins at the links' CGs (LNCS). Some of the linkage parameters are normally expressed in such coordinate systems, but some are not, and so must be converted. The kinematic data should all be computed in the global system or in parallel, **nonrotating**, local systems placed at the CGs of individual links. Any external forces on the links must also be defined in the global system.

However, the position vectors that define intralink locations, such as the pin joints versus the *CG*, or which locate points of application of external forces versus the *CG* are defined in local, **rotating** coordinate systems embedded in the links (LRCS). Thus these position vectors must be redefined in a **nonrotating**, parallel system before being used in the matrix. An example of this is vector $\mathbf{R}_p$, which was initially defined as 3 in at 100° in link 3's embedded, **rotating** coordinate system. Note in the example above that its cartesian coordinates for use in the equations were calculated after adding the current value of θ_3 to its angle. This redefined $\mathbf{R}_p$ as 3 in at 120.92° in the **nonrotating** local system. The same was done for position vectors $\mathbf{R}_{12}$, $\mathbf{R}_{32}$, $\mathbf{R}_{23}$, $\mathbf{R}_{43}$, $\mathbf{R}_{34}$, and $\mathbf{R}_{14}$. In each case the **intralink angle** of these vectors (which is independent of linkage position) was added to the current link angle to obtain its position in the *xy* system at the link's *CG*. The proper definition of these position vector components is critical to the solution, and it is very easy to make errors in defining them.

To further confuse things, even though the position vector $\mathbf{R}_p$ is initially measured in the link's embedded, rotating coordinate system, the force $\mathbf{F}_p$, which it locates, is not. The force $\mathbf{F}_p$ is not part of the link, as is $\mathbf{R}_p$, but rather is part of the external world, so it is defined in the global system.

11.5 FORCE ANALYSIS OF A FOURBAR SLIDER-CRANK LINKAGE

The approach taken for the pin-jointed fourbar is equally valid for a fourbar slider-crank linkage. The principal difference will be that the slider block will have no angular acceleration. Figure 11-4 shows a fourbar slider-crank with an external force on the slider block, link 4. This is representative of the mechanism used extensively in piston pumps and internal combustion engines. We wish to determine the forces at the joints and the driving torque needed on the crank to provide the specified accelerations. A kinematic analysis must have previously been done in order to determine all position, velocity, and acceleration information for the positions being analyzed. Equations 11.1 are written for each link. For link 2:

$$\begin{aligned} F_{12_x} + F_{32_x} &= m_2 a_{G2_x} \\ F_{12_y} + F_{32_y} &= m_2 a_{G2_y} \\ T_{12} + (R_{12_x} F_{12_y} - R_{12_y} F_{12_x}) + (R_{32_x} F_{32_y} - R_{32_y} F_{32_x}) &= I_{G2} \alpha_2 \end{aligned} \quad (11.10a)$$

This is identical to equation 11.8a for the "pure" fourbar linkage. For link 3:

$$\begin{aligned} F_{43_x} - F_{32_x} &= m_3 a_{G3_x} \\ F_{43_y} - F_{32_y} &= m_3 a_{G3_y} \\ (R_{43_x} F_{43_y} - R_{43_y} F_{43_x}) - (R_{23_x} F_{32_y} - R_{23_y} F_{32_x}) &= I_{G3} \alpha_3 \end{aligned} \quad (11.10b)$$

This is similar to equation 11.8b, lacking only the terms involving $\mathbf{F}_p$ since there is no external force shown acting on link 3 of our example slider-crank. For link 4:

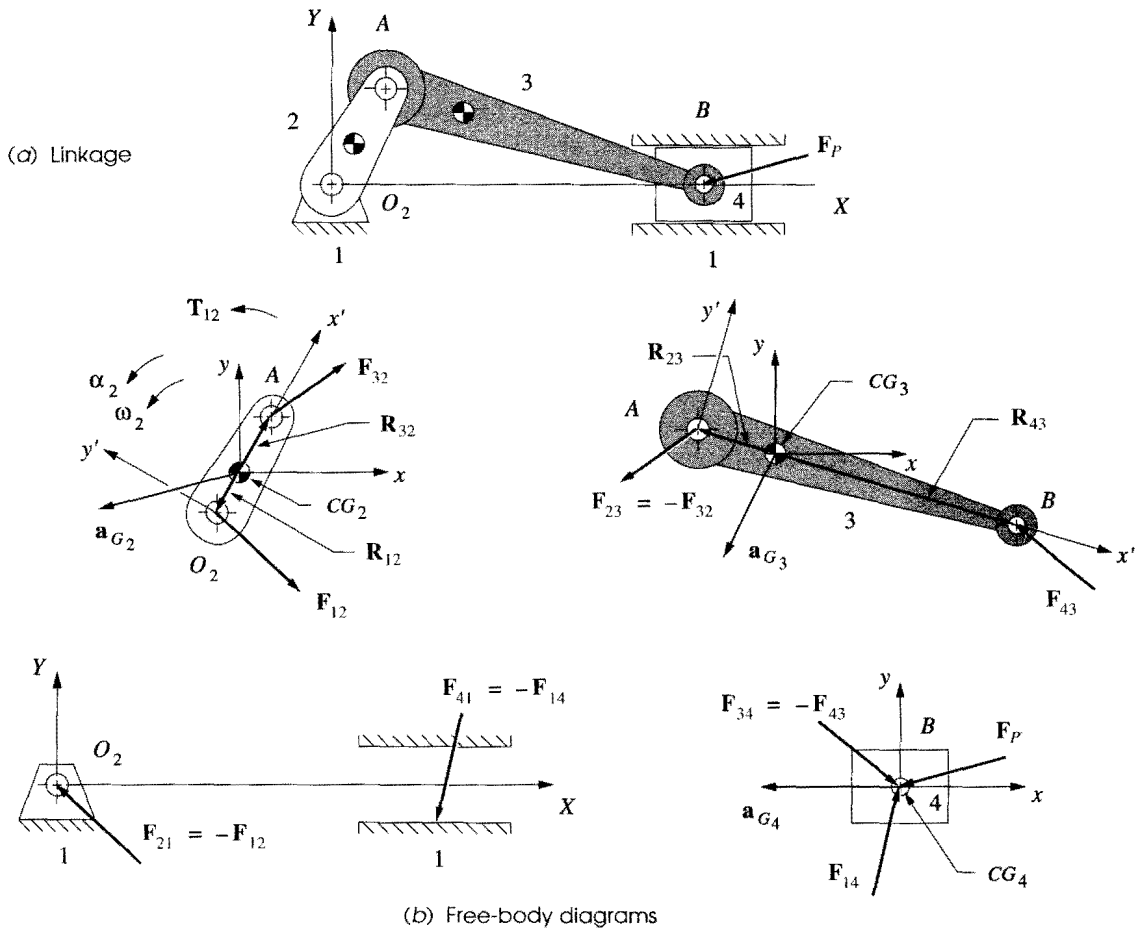


FIGURE 11-4

Dynamic force analysis of the fourbar slider-crank linkage

$$\begin{aligned}
 F_{14x} - F_{43x} + F_{p_x} &= m_4 a_{G4x} \\
 F_{14y} - F_{43y} + F_{p_y} &= m_4 a_{G4y}
 \end{aligned}
 \quad (11.10c)$$

$$\left(R_{14x} F_{14y} - R_{14y} F_{14x} \right) - \left(R_{34x} F_{43y} - R_{34y} F_{43x} \right) + \left(R_{p_x} F_{p_y} - R_{p_y} F_{p_x} \right) = I_{G4} \alpha_4$$

These contain the external force $\mathbf{F}_p$ shown acting on link 4.

For the inversion of the slider-crank shown, the slider block, or piston, is in pure translation against the stationary ground plane; thus it can have no angular acceleration or angular velocity. Also, the position vectors in the torque equation (equation 11.10c) are all zero as the force $\mathbf{F}_p$ acts at the CG. Thus the torque equation for link 4 (third expression in equation 11.10c) is zero for this inversion of the slider-crank linkage. Its linear acceleration also has no y component.

$$\alpha_4 = 0, \quad a_{G_4y} = 0 \quad (11.10d)$$

The only x directed force that can exist at the interface between links 4 and 1 is friction. Assuming coulomb friction, the x component can be expressed in terms of the y component of force at this interface. We can write a relation for the friction force f at that interface such as $f = \pm\mu N$, where $\pm\mu$ is a known coefficient of friction. The plus and minus signs on the coefficient of friction are to recognize the fact that the friction force always opposes motion. The kinematic analysis will provide the velocity of the link at the sliding joint. The sign of μ will always be the opposite of the sign of this velocity.

$$F_{14x} = \pm\mu F_{14y} \quad (11.10e)$$

Substituting equations 11.10d and 11.10e into the reduced equation 11.10c yields:

$$\begin{aligned} \pm\mu F_{14y} - F_{43x} + F_{Px} &= m_4 a_{G_4x} \\ F_{14y} - F_{43y} + F_{Py} &= 0 \end{aligned} \quad (11.10f)$$

This last substitution has reduced the unknowns to eight, F_{12x} , F_{12y} , F_{32x} , F_{32y} , F_{43x} , F_{43y} , F_{14y} , and T_{12} ; thus we need only eight equations. We can now use the eight equations in 11.10a, b, and f to assemble the matrices for solution.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -R_{12y} & R_{12x} & -R_{32y} & R_{32x} & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & R_{23y} & -R_{23x} & -R_{43y} & R_{43x} & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & \pm\mu & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} F_{12x} \\ F_{12y} \\ F_{32x} \\ F_{32y} \\ F_{43x} \\ F_{43y} \\ F_{14y} \\ T_{12} \end{bmatrix} = \quad (11.10g)$$

$$\begin{bmatrix} m_2 a_{G_2x} \\ m_2 a_{G_2y} \\ I_{G_2} \alpha_2 \\ m_3 a_{G_3x} \\ m_3 a_{G_3y} \\ I_{G_3} \alpha_3 \\ m_4 a_{G_4x} - F_{Px} \\ -F_{Py} \end{bmatrix}$$

Solution of this matrix equation 11.10g plus equation 11.10e will yield complete dynamic force information for the fourbar slider-crank linkage.

11.6 FORCE ANALYSIS OF THE INVERTED SLIDER-CRANK

Another inversion of the fourbar slider-crank was also analyzed kinematically in Part I. It is shown in Figure 11-5. Link 4 does have an angular acceleration in this inversion. In fact, it must have the same angle, angular velocity, and angular acceleration as link 3 because they are rotationally coupled by the sliding joint. We wish to determine the forces at all pin joints and at the sliding joint as well as the driving torque needed to create the desired accelerations. Each link's joints are located by position vectors referenced to nonrotating local xy coordinate systems at each link's CG as before. The sliding joint is located by the position vector $\mathbf{R}_{43}$ to the center of the slider, point B . The instantaneous position of point B was determined from the kinematic analysis as length b referenced to instant center I_{23} (point A). See Sections 4.7 (p. 161), 6.7 (p. 276), and 7.3 (p. 315) to review the position, velocity, and acceleration analysis of this mechanism. Recall that this mechanism has a nonzero Coriolis component of acceleration. The force between link 3 and link 4 within the sliding joint is distributed along the unspecified length of the slider block. For this analysis the distributed force can be modeled as a force concentrated at point B within the sliding joint. We will neglect friction in this example.

The equations for links 2 and 3 are identical to those for the noninverted slider-crank (Equations 11.10a and b). The equations for link 4 are the same as equations 11.10c except for the absence of the terms involving $\mathbf{F}_p$ since no external force is shown acting on link 4 in this example. The slider joint can only transmit force from link 3 to link 4 or vice versa along a line perpendicular to the axis of slip. This line is called the axis of transmission. In order to guarantee that the force $\mathbf{F}_{34}$ or $\mathbf{F}_{43}$ is always perpendicular to the axis of slip, we can write the following relation:

$$\hat{\mathbf{u}} \cdot \mathbf{F}_{43} = 0 \quad (11.11a)$$

which expands to:

$$u_x F_{43x} + u_y F_{43y} = 0 \quad (11.11b)$$

The dot product of two vectors will be zero when the vectors are mutually perpendicular. The unit vector $\mathbf{u}$ is in the direction of link 3 which is defined from the kinematic analysis as θ_3 .

$$u_x = \cos \theta_3, \quad u_y = \sin \theta_3 \quad (11.11c)$$

Equation 11.11 provides a tenth equation, but we have only nine unknowns, F_{12x} , F_{12y} , F_{32x} , F_{32y} , F_{43x} , F_{43y} , F_{14x} , F_{14y} , and T_{12} , so one of our equations is redundant. Since we must include equation 11.11, we will combine the torque equations for links 3 and 4 rewritten here in vector form and without the external force $\mathbf{F}_p$.

$$(\mathbf{R}_{43} \times \mathbf{F}_{43}) - (\mathbf{R}_{23} \times \mathbf{F}_{32}) = I_{G_3} \alpha_3 = I_{G_3} \alpha_4 \quad (11.12a)$$

$$(\mathbf{R}_{14} \times \mathbf{F}_{14}) - (\mathbf{R}_{34} \times \mathbf{F}_{43}) = I_{G_4} \alpha_4$$

Note that the angular acceleration of link 3 is the same as that of link 4 in this linkage. Adding these equations gives:

$$(\mathbf{R}_{43} \times \mathbf{F}_{43}) - (\mathbf{R}_{23} \times \mathbf{F}_{32}) + (\mathbf{R}_{14} \times \mathbf{F}_{14}) - (\mathbf{R}_{34} \times \mathbf{F}_{43}) = (I_{G_3} + I_{G_4}) \alpha_4 \quad (11.12b)$$

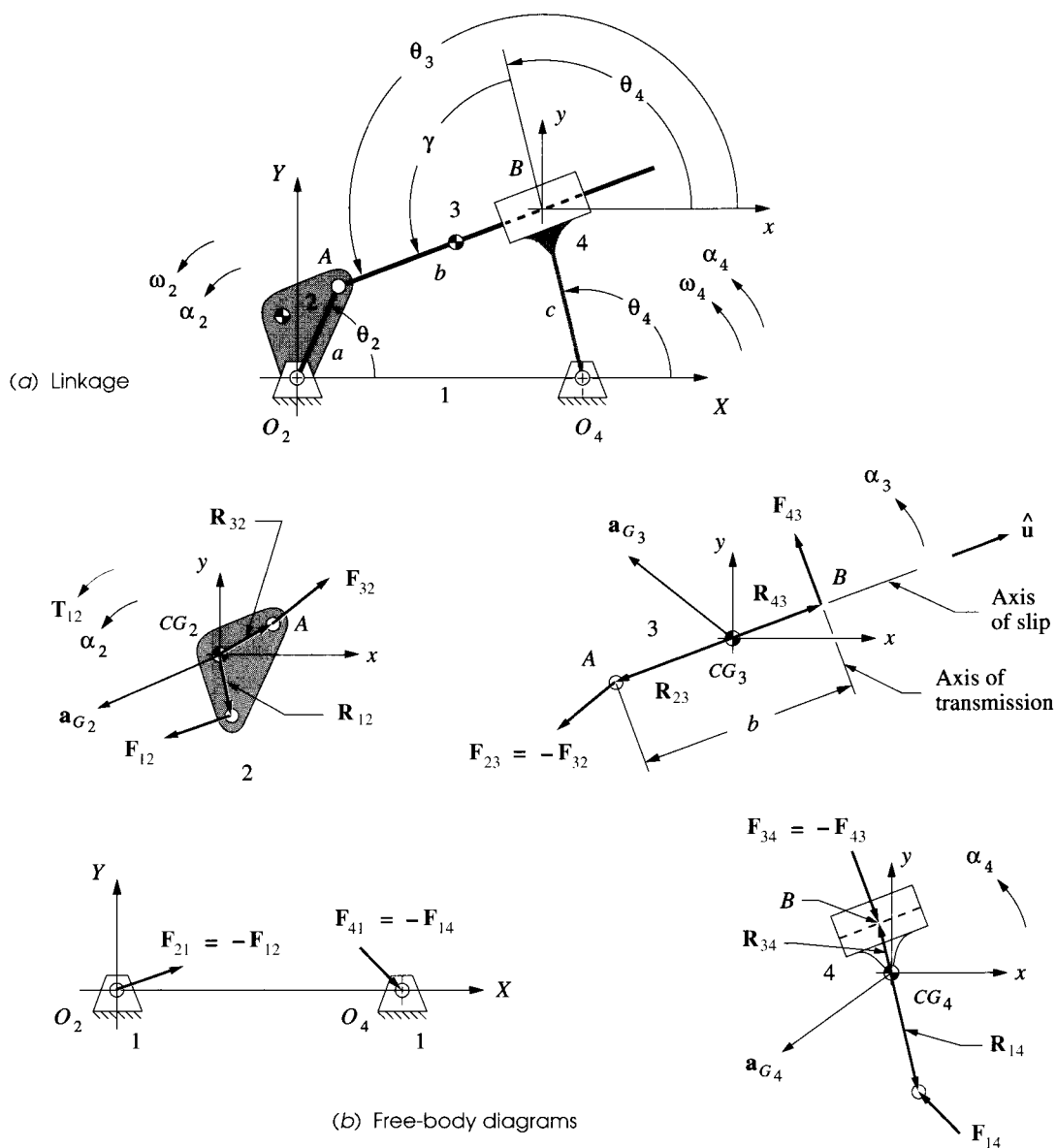


FIGURE 11-5

Dynamic forces in the inverted slider-crank fourbar linkage

Expanding and collecting terms:

$$\begin{aligned}
 & (R_{43x} - R_{34x})F_{43y} + (R_{34y} - R_{43y})F_{43x} - R_{23x}F_{32y} \\
 & + R_{23y}F_{32x} + R_{14x}F_{14y} - R_{14y}F_{14x} = (I_{G_3} + I_{G_4})\alpha_4
 \end{aligned}
 \quad (11.12c)$$

Equations 11.10a, 11.11b, 11.12c, and the four force equations from equations 11.10b and 11.10c (excluding the external force $\mathbf{F}_P$) give us nine equations in the nine unknowns which we can put in matrix form for solution.

$$\begin{bmatrix}
 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
 -R_{12y} & R_{12x} & -R_{32y} & R_{32x} & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & R_{23y} & -R_{23x} & (R_{34y} - R_{43y}) & (R_{43x} - R_{34x}) & -R_{14y} & R_{14x} & 0 \\
 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & u_x & u_y & 0 & 0 & 0
 \end{bmatrix} \times$$

$$\begin{bmatrix}
 F_{12x} \\
 F_{12y} \\
 F_{32x} \\
 F_{32y} \\
 F_{43x} \\
 F_{43y} \\
 F_{14x} \\
 F_{14y} \\
 T_{12}
 \end{bmatrix} = \begin{bmatrix}
 m_2 a_{G_2x} \\
 m_2 a_{G_2y} \\
 I_{G_2} \alpha_2 \\
 m_3 a_{G_3x} \\
 m_3 a_{G_3y} \\
 (I_{G_3} + I_{G_4}) \alpha_4 \\
 m_4 a_{G_4x} \\
 m_4 a_{G_4y} \\
 0
 \end{bmatrix} \quad (11.13)$$

11.7 FORCE ANALYSIS—LINKAGES WITH MORE THAN FOUR BARS

This matrix method of force analysis can easily be extended to more complex assemblages of links. The equations for each link are of the same form. We can create a more general notation for equations 11.1 to apply them to any assembly of n pin-connected links. Let j represent any link in the assembly. Let $i = j - 1$ be the previous link in the chain and $k = j + 1$ be the next link in the chain; then, using the vector form of equations 11.1:

$$\mathbf{F}_{ij} + \mathbf{F}_{jk} + \sum \mathbf{F}_{extj} = m_j \mathbf{a}_{G_j} \quad (11.14a)$$

$$(\mathbf{R}_{ij} \times \mathbf{F}_{ij}) + (\mathbf{R}_{jk} \times \mathbf{F}_{jk}) + \sum \mathbf{T}_j + (\mathbf{R}_{extj} \times \sum \mathbf{F}_{extj}) = I_{G_j} \alpha_j \quad (11.14b)$$

where:

$$j = 2, 3, \dots, n; \quad i = j - 1; \quad k = j + 1, j \neq n; \quad \text{if } j = n, k = 1$$

$$\text{and} \quad \mathbf{F}_{ji} = -\mathbf{F}_{ij}; \quad \mathbf{F}_{kj} = -\mathbf{F}_{jk} \quad (11.14c)$$

The sum of forces vector equation 11.14a can be broken into its two x and y component equations and then applied, along with the sum of torques equation 11.14b, to each of the links in the chain to create the set of simultaneous equations for solution. Any link

may have external forces and/or external torques applied to it. All will have pin forces. Since the n th link in a closed chain connects to the first link, the value of k for the n th link is set to 1. In order to reduce the number of variables to a tractable quantity, substitute the negative reaction forces from equation 11.14c where necessary as was done in the examples above. When sliding joints are present, it will be necessary to add constraints on the allowable directions of forces at those joints as was done in the inverted slider-crank example above.

11.8 SHAKING FORCES AND SHAKING TORQUE

It is usually of interest to know the net effect of the dynamic forces as felt on the ground plane as this can set up vibrations in the structure which supports the machine. For our simple examples of three- and fourbar linkages, there are only two points at which the dynamic forces can be delivered to link 1, the ground plane. More complicated mechanisms will have more joints with the ground plane. The forces delivered by the moving links to the ground at the fixed pivots O_2 and O_4 are designated $\mathbf{F}_{21}$ and $\mathbf{F}_{41}$ by our subscript convention as defined in Section 11.1 (p. 521). Since we chose to solve for $\mathbf{F}_{12}$ and $\mathbf{F}_{14}$ in our solutions, we simply negate those forces to obtain their equal and opposite counterparts (see also equation 11.5, p. 527).

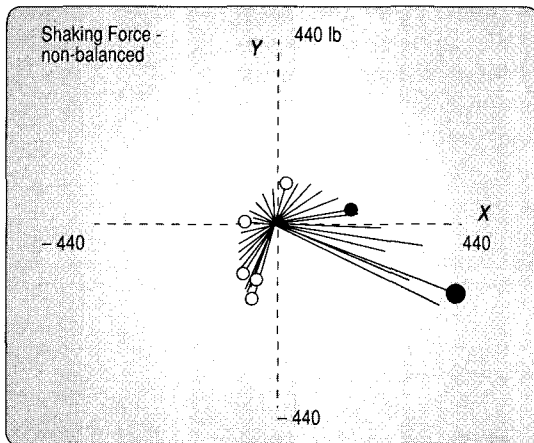
$$\mathbf{F}_{21} = -\mathbf{F}_{12} \qquad \mathbf{F}_{41} = -\mathbf{F}_{14} \qquad (11.15a)$$

The *sum of all the forces acting on the ground plane* is called the **shaking force** ($\mathbf{F}_s$) as shown in Figure 11-6.* In these simple examples it is equal to:

$$\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41} \qquad (11.15b)$$

The *reaction torque felt by the ground plane* is also called the **shaking torque** ($\mathbf{T}_s$) as shown in Figure 11-7.* This is the negative of the source torque $\mathbf{T}_{12}$ which is delivered to the driving link from the ground.

* The FOURBAR disk file (F11-06.4br) that generated the plots in Figures 11-6 and 11-7 may be opened in that program to see more detail on the linkage's dynamics.



Link No.	Length in	Mass Units	Inertia Units	CG Posit	at Deg	Ext. Force lb	at Deg
1	5.5						
2	2.0	.002	.004	1.0	0		
3	6.0	.030	.060	2.5	30	12	270
4	3.0	.010	.020	1.5	0	60	-45

Coupler pt. = 3 in @ 45°

Open/Crossed = open

Ext. Force 3 acts at 5 in @ 30° vs. CG of Link 3

Ext. Force 4 acts at 5 in @ 90° vs. CG of Link 4

Ext. Torque 3 = -20 lb-in

Ext. Torque 4 = 25 lb-in

Start Alpha2 = 0 rad/sec²

Start Omega2 = 50 rad/sec

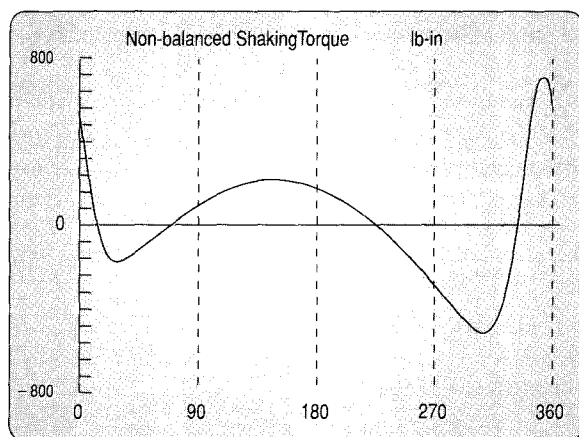
Start Theta2 = 0°

Final Theta2 = 360°

Delta Theta2 = 10°

FIGURE 11-6

Linkage data and polar plot of shaking force for an unbalanced crank-rocker fourbar linkage from program FOURBAR



Link No.	Length in	Mass Units	Inertia Units	CG Posit	at Deg	Ext. Force lb	at Deg
1	5.5						
2	2.0	.002	.004	1.0	0		
3	6.0	.030	.060	2.5	30	12	270
4	3.0	.010	.020	1.5	0	60	-45

Coupler pt. = 3 in @ 45°

Open/Crossed = open

Ext. Force 3 acts at 5 in @ 30° vs. CG of Link 3

Ext. Force 4 acts at 5 in @ 90° vs. CG of Link 4

Ext. Torque 3 = -20 lb-in

Ext. Torque 4 = 25 lb-in

Start Alpha2 = 0 rad/sec²

Start Omega2 = 50 rad/sec

Start Theta2 = 0°

Final Theta2 = 360°

Delta Theta2 = 2°

FIGURE 11-7

Linkage data and shaking torque curve for an unbalanced crank-rocker fourbar linkage from program FOURBAR

$$T_s = T_{21} = -T_{12} \quad (11.15c)$$

The shaking force will tend to move the ground plane back and forth, and the shaking torque will tend to rock the ground plane about the driveline axis. Both will cause vibrations. We are usually looking to minimize the effects of the shaking forces and shaking torques on the frame. This can sometimes be done by balancing, sometimes by the addition of a flywheel to the system, and sometimes by shock mounting the frame to isolate the vibrations from the rest of the assembly. Most often we will use a combination of all three approaches. We will investigate some of these techniques shortly.

11.9 PROGRAM FOURBAR

The matrix methods introduced in the preceding sections all provide force and torque information for one position of the linkage assembly as defined by its kinematic and geometric parameters. To do a complete force analysis for multiple positions of a machine requires that these computations be repeated with new input data for each position. A computer program is the obvious way to accomplish this. Program FOURBAR, on the enclosed CD-ROM, computes the kinematic parameters for any fourbar linkage over changes in time or driver (crank) angle plus the forces and torques concomitant with the linkage kinematics and link geometry. Examples of its output are shown in Figures 11-6 and 11-7. Please refer to Appendix A for information on the use of program FOURBAR.

11.10 LINKAGE FORCE ANALYSIS BY ENERGY METHODS

In Section 10.13 (p. 515) the method of virtual work was presented. We will now use that approach to solve the linkage from Example 11-3 as a check on its solution by the newtonian method used above. The kinematic data given in Example 11-3 did not include information on the angular velocities of all the links, the linear velocities of the centers of gravities of the links, and the linear velocity of the point *P* of application of

the external force on link 3. Velocity data were not needed for the newtonian solution but are for the virtual work approach and are detailed below. Equation 10.26a is repeated here.

$$\sum_{k=2}^n \mathbf{F}_k \cdot \mathbf{v}_k + \sum_{k=2}^n \mathbf{T}_k \cdot \boldsymbol{\omega}_k = \sum_{k=2}^n m_k \mathbf{a}_k \cdot \mathbf{v}_k + \sum_{k=2}^n I_k \alpha_k \cdot \boldsymbol{\omega}_k \quad (11.16a)$$

Expanding the summations, still in vector form:

$$\begin{aligned} & (\mathbf{F}_{P_3} \cdot \mathbf{v}_{P_3} + \mathbf{F}_{P_4} \cdot \mathbf{v}_{P_4}) + (\mathbf{T}_{12} \cdot \boldsymbol{\omega}_2 + \mathbf{T}_3 \cdot \boldsymbol{\omega}_3 + \mathbf{T}_4 \cdot \boldsymbol{\omega}_4) = \\ & (m_2 \mathbf{a}_{G_2} \cdot \mathbf{v}_{G_2} + m_3 \mathbf{a}_{G_3} \cdot \mathbf{v}_{G_3} + m_4 \mathbf{a}_{G_4} \cdot \mathbf{v}_{G_4}) \\ & + (I_{G_2} \alpha_2 \cdot \boldsymbol{\omega}_2 + I_{G_3} \alpha_3 \cdot \boldsymbol{\omega}_3 + I_{G_4} \alpha_4 \cdot \boldsymbol{\omega}_4) \end{aligned} \quad (11.16b)$$

Expanding the dot products to create a scalar equation:

$$\begin{aligned} & (F_{P_3x} V_{P_3x} + F_{P_3y} V_{P_3y}) + (F_{P_4x} V_{P_4x} + F_{P_4y} V_{P_4y}) + (T_{12} \omega_2 + T_3 \omega_3 + T_4 \omega_4) = \\ & m_2 (a_{G_2x} V_{G_2x} + a_{G_2y} V_{G_2y}) + m_3 (a_{G_3x} V_{G_3x} + a_{G_3y} V_{G_3y}) \\ & + m_4 (a_{G_4x} V_{G_4x} + a_{G_4y} V_{G_4y}) + (I_{G_2} \alpha_2 \omega_2 + I_{G_3} \alpha_3 \omega_3 + I_{G_4} \alpha_4 \omega_4) \end{aligned} \quad (11.16c)$$



EXAMPLE 11-4

Analysis of a Fourbar Linkage by the Method of Virtual Work. (See Figure 11-3, p. 532.)

Given: The 5-in-long crank (link 2) shown weighs 1.5 lb. Its *CG* is at 3 in at $+30^\circ$ from the line of centers. Its mass moment of inertia about its *CG* is 0.4 lb-in-sec². Its kinematic data are:

θ_2 deg	ω_2 rad/sec	α_2 rad/sec ²	V_{G_2} in/sec
60	25	-40	75 @ 180°

The coupler (link 3) is 15 in long and weighs 7.7 lb. Its *CG* is at 9 in at 45° off the line of centers. Its mass moment of inertia about its *CG* is 1.5 lb-in-sec². Its kinematic data are:

θ_3 deg	ω_3 rad/sec	α_3 rad/sec ²	V_{G_3} in/sec
20.92	-5.87	120.9	72.66 @ 145.7°

There is an external force on link 3 of 80 lb at 330° , applied at point *P* which is located 3 in @ 100° from the *CG* of link 3. The linear velocity of that point is 67.2 in/sec at 131.94° .

The rocker (link 4) is 10 in long and weighs 5.8 lb. Its *CG* is at 5 in at 0° off the line of centers. Its mass moment of inertia about its *CG* is 0.8 lb-in-sec². Its kinematic data are:

θ_4 deg	ω_4 rad/sec	α_4 rad/sec ²	V_{G_4} in/sec
104.41	7.93	276.29	39.66 @ 194.41°

There is an external torque on link 4 of 120 lb-in. The ground link is 19 in long.

Find: The driving torque T_{12} needed to maintain motion with the given acceleration for this instantaneous position of the link.

Solution:

- 1 The torque, angular velocity, and angular acceleration vectors in this two-dimensional problem are all directed along the Z axis, so their dot products each have only one term. Note that in this particular example there is no force $\mathbf{F}_{P4}$ and no torque $\mathbf{T}_3$.

- 2 The cartesian coordinates of the acceleration data were calculated in Example 11-3 above.

$$\begin{aligned} \mathbf{a}_{G_2} &= 1878.84 \text{ @ } \angle 273.66^\circ; & a_{G_{2x}} &= 119.94, & a_{G_{2y}} &= -1875.01 \\ \mathbf{a}_{G_3} &= 3646.10 \text{ @ } \angle 226.51^\circ; & a_{G_{3x}} &= -2509.35, & a_{G_{3y}} &= -2645.23 \\ \mathbf{a}_{G_4} &= 1416.80 \text{ @ } \angle 207.24^\circ; & a_{G_{4x}} &= -1259.67, & a_{G_{4y}} &= -648.50 \end{aligned} \quad (a)$$

- 3 The x and y components of the external force at P in the global coordinate system were also calculated in Example 11-3:

$$\mathbf{F}_P = 80 \text{ @ } \angle 330^\circ; \quad F_{P_x} = 69.28, \quad F_{P_y} = -40.00 \quad (b)$$

- 4 Converting the velocity data for this example to cartesian coordinates:

$$\begin{aligned} \mathbf{V}_{G_2} &= 75 \text{ @ } \angle 180^\circ; & V_{G_{2x}} &= -75, & V_{G_{2y}} &= 0 \\ \mathbf{V}_{G_3} &= 72.66 \text{ @ } \angle 145.70^\circ; & V_{G_{3x}} &= -60.02, & V_{G_{3y}} &= 40.95 \\ \mathbf{V}_{G_4} &= 39.66 \text{ @ } \angle 194.41^\circ; & V_{G_{4x}} &= -38.41, & V_{G_{4y}} &= -9.87 \\ \mathbf{V}_P &= 67.20 \text{ @ } \angle 131.94^\circ; & V_{P_x} &= -44.91, & V_{P_y} &= 49.99 \end{aligned} \quad (c)$$

- 5 Substituting the example data into equation 11.16c:

$$\begin{aligned} &[(69.28)(-44.91) + (-40)(49.99)] + [0] + [25T_{12} + (0) + (120)(7.93)] = \\ &\frac{1.5}{386}[(119.94)(-75) + (-1875.01)(0)] \\ &+ \frac{7.7}{386}[(-2509.35)(-60.02) + (-2645.23)(40.95)] \\ &+ \frac{5.8}{386}[(-1259.67)(-38.41) + (-648.50)(-9.87)] \\ &+ [(0.4)(-40)(25) + (1.5)(120.9)(-5.87) + (0.8)(276.29)(7.93)] \end{aligned} \quad (d)$$

- 6 The only unknown in this equation is the input torque T_{12} which calculates to:

$$\mathbf{T}_{12} = 243.2 \hat{\mathbf{k}} \quad (e)$$

which is the same as the answer obtained in Example 11-3.

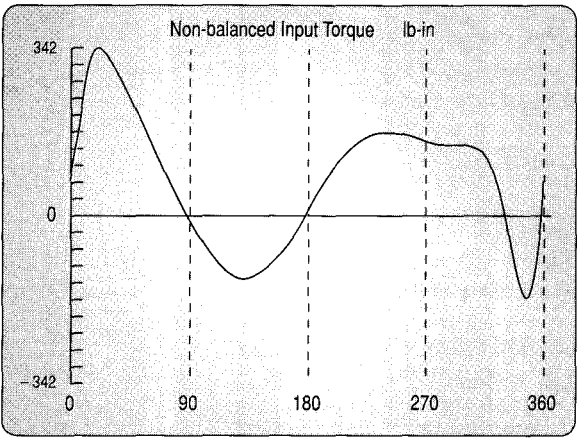
This method of virtual work is useful if a quick answer is needed for the input torque, but it does not give any information about the joint forces.

11.11 CONTROLLING INPUT TORQUE-FLYWHEELS

The typically large variation in accelerations within a mechanism can cause significant oscillations in the torque required to drive it at a constant or near constant speed. The peak torques needed may be so high as to require an overly large motor to deliver them. However, the average torque over the cycle, due mainly to losses and external work done, may often be much smaller than the peak torque. We would like to provide some means to smooth out these oscillations in torque during the cycle. This will allow us to size the motor to deliver the average torque rather than the peak torque. One convenient and relatively inexpensive means to this end is the addition of a flywheel to the system.

TORQUE VARIATION Figure 11-8 shows the variation in the input torque for a crank-rocker fourbar linkage over one full revolution of the drive crank. It is running at a constant angular velocity of 50 rad/sec. The torque varies a great deal within one cycle of the mechanism, going from a positive peak of 341.7 lb-in to a negative peak of -166.41b-in. The average value of this torque over the cycle is only 70.21b-in, being due to the *external work done plus losses*. This linkage has only a 12-lb external force applied to link 3 at the CG and a 25 lb-in external torque applied to link 4. These small external loads cannot account for the large variation in input torque required to maintain constant crank speed. What then is the explanation? The large variations in torque are evidence of the kinetic energy that is stored in the links as they move. We can think of the positive pulses of torque as representing energy delivered by the driver (motor) and stored temporarily in the moving links, and the negative pulses of torque as energy attempting to return from the links to the driver. Unfortunately most motors are designed to deliver energy but not to take it back. Thus the "returned energy" has no place to go.

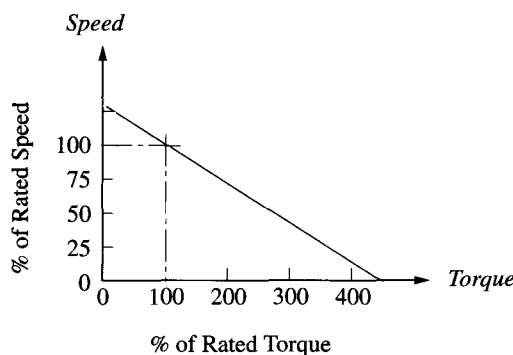
Figure 11-9 shows the speed torque characteristic of a permanent magnet (PM) DC electric motor. Other types of motors will have differently shaped functions that relate motor speed to torque as shown in Figure 2-32 and 2-33 (pp. 62-63), but all drivers



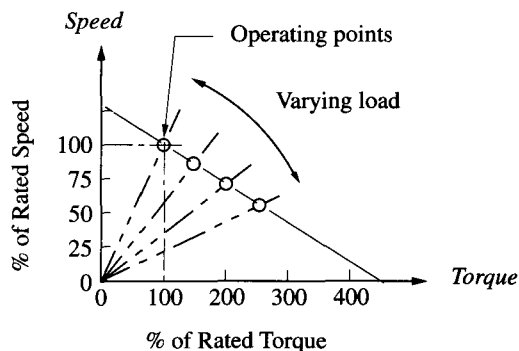
Link No.	Length in	Mass Units	Inertia Units	CG Posit	at Deg	Ext. Force lb	at Deg
1	5.5						
2	2.0	.002	.004	1.0	0		
3	6.0	.030	.060	2.5	30	12	270
4	3.0	.010	.020	1.5	0	60	-45

Coupler pt. = 3 in @ 45°
 Open/Crossed = open
 Ext. Force 3 acts at 5 in @ 30° vs. CG of Link 3
 Ext. Force 4 acts at 5 in @ 90° vs. CG of Link 4
 Ext. Torque 3 = - 20 lb-in
 Ext. Torque 4 = 25 lb-in
 Start Alpha2 = 0 rad/sec²
 Start Omega2 = 50 rad/sec
 Start Theta2 = 0°
 Final Theta2 = 360°
 Delta Theta2 = 2°

FIGURE 11-8
 Linkage data and input torque curve for an unbalanced crank-rocker fourbar linkage from program FOURBAR



(a) Speed-torque characteristic of a PM electric motor



(b) Load lines superposed on speed-torque curve

FIGURE 11-9

DC permanent magnet (PM) electric motor's typical speed-torque characteristic

(sources) will have some such characteristic curve. As the torque demands on the motor change, the motor's speed must also change according to its inherent characteristic. This means that the torque curve being demanded in Figure 11-8 will be very difficult for a standard motor to deliver without drastic changes in its speed.

The computation of the torque curve in Figure 11-8 was made on the assumption that the crank (thus the motor) speed was a constant value. All the kinematic data used in the force and torque calculation was generated on that basis. With the torque variation shown we would have to use a large-horsepower motor to provide the power required to reach that peak torque at the design speed:

$$\text{Power} = \text{torque} \times \text{angular velocity}$$

$$\text{Peak power} = 341.7 \text{ lb} \cdot \text{in} \times 50 \frac{\text{rad}}{\text{sec}} = 17,085 \frac{\text{in} \cdot \text{lb}}{\text{sec}} = 2.59 \text{ hp}$$

The power needed to supply the average torque is much smaller.

$$\text{Average power} = 70.2 \text{ lb} \cdot \text{in} \times 50 \frac{\text{rad}}{\text{sec}} = 3,510 \frac{\text{in} \cdot \text{lb}}{\text{sec}} = 0.53 \text{ hp}$$

It would be extremely inefficient to specify a motor based on the peak demand of the system, as most of the time it will be underutilized. We need something in the system which is capable of storing kinetic energy. One such kinetic energy storage device is called a **flywheel**.

FLYWHEEL ENERGY Figure 11-10 shows a **flywheel**, designed as a flat circular disk, attached to a motor shaft which might also be the driveshaft for the crank of our linkage. The motor supplies a torque magnitude T_M which we would like to be as constant as possible, i.e., to be equal to the average torque T_{avg} . The load (our linkage), on the other side of the flywheel, demands a torque T_L which is time varying as shown in Figure 11-8. The kinetic energy in a rotating system is:

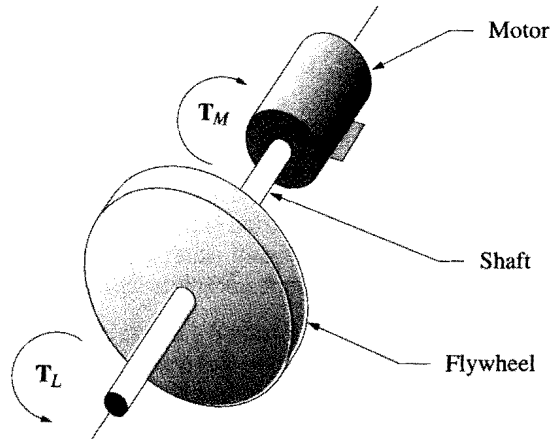


FIGURE 11-10

Flywheel on a driveshaft

$$E = \frac{1}{2} I \omega^2 \quad (11.17)$$

where I is the moment of inertia of all rotating mass on the shaft. This includes the I of the motor rotor and of the linkage crank plus that of the flywheel. We want to determine how much I we need to add in the form of a flywheel to reduce the speed variation of the shaft to an acceptable level. We begin by writing Newton's law for the free-body diagram in Figure 11-10.

$$\sum T = I \alpha$$

$$T_L - T_M = I \alpha$$

but we want :

$$T_M = T_{avg}$$

so :

$$T_L - T_{avg} = I \alpha \quad (11.18a)$$

substituting :

$$\alpha = \frac{d\omega}{dt} = \frac{d\omega}{dt} \left(\frac{d\theta}{d\theta} \right) = \omega \frac{d\omega}{d\theta}$$

gives :

$$T_L - T_{avg} = I \omega \frac{d\omega}{d\theta}$$

$$(T_L - T_{avg}) d\theta = I \omega d\omega \quad (11.18b)$$

and integrating:

$$\int_{\theta @ \omega_{min}}^{\theta @ \omega_{max}} (T_L - T_{avg}) d\theta = \int_{\omega_{min}}^{\omega_{max}} I \omega d\omega$$

(11.18c)

$$\int_{\theta @ \omega_{min}}^{\theta @ \omega_{max}} (T_L - T_{avg}) d\theta = \frac{1}{2} I (\omega_{max}^2 - \omega_{min}^2)$$

The left side of this expression represents the change in energy E between the maximum and minimum shaft ω 's and is equal to the *area under the torque-time diagram** (Figures 11-8, p. 548, and 11-11) between those extreme values of ω . The right side of equation 11.18c is the change in energy stored in the flywheel. The only way we can extract energy from the flywheel is to slow it down as shown in equation 11.17. Adding energy will speed it up. Thus it is impossible to obtain exactly constant shaft velocity in the face of changing energy demands by the load. The best we can do is to minimize the speed variation ($\omega_{max} - \omega_{min}$) by providing a flywheel with sufficiently large I .

EXAMPLE 11-5

Determining the Energy Variation in a Torque-Time Function.

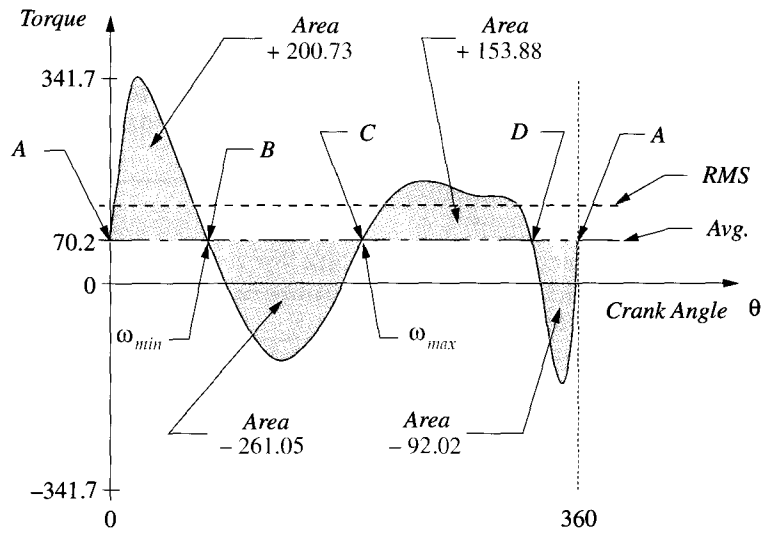
Given: An input torque-time function which varies over its cycle. Figure 11-11 shows the input torque curve from Figure 11-8. The torque is varying during the 360° cycle about its average value.

Find: The total energy variation over one cycle.

Solution:

- 1 Calculate the average value of the torque-time function over one cycle, which in this case is 70.2 lb-in. (Note that in some cases the average value may be zero.)
- 2 Note that the *integration on the left side of equation 11.18c is done with respect to the average line of the torque function, not with respect to the θ axis.* (From the definition of the

* There is often confusion between torque and energy because they appear to have the same units of *lb-in* (*in-lb*) or *N-m* (*m-N*). This leads some students to think that they are the same quantity, but they are not. Torque $\neq$ energy. The **integral** of torque with respect to angle, measured in radians, is equal to energy. This integral has the units of *in-lb-rad*. The radian term is usually omitted since it is in fact unity. Power in a rotating system is equal to torque $\times$ angular velocity (measured in *rad/sec*), and the power units are then (*in-lb-rad*)/*sec*. When power is integrated versus time to get energy, the resulting units are *in-lb-rad*, the same as the integral of torque versus angle. The radians are again usually dropped, contributing to the confusion.



Areas of torque pulses in order over one cycle		
Order	Neg Area	Pos Area
1	- 261.05	200.73
2	- 92.02	153.88

Energy units are lb-in-rad

FIGURE 11-11

Integrating the pulses above and below the average value in the input torque function

average, the sum of positive area above an average line is equal to the sum of negative area below that line.) The integration limits in equation 11.18 are from the shaft angle θ at which the shaft (j) is a minimum to the shaft angle θ at which (j) is a maximum.

- 3 The minimum (j) will occur after the maximum positive energy has been delivered from the motor to the load, i.e., at a point (8) where the summation of positive energy (area) in the torque pulses is at its largest positive value.
- 4 The maximum (j) will occur after the maximum negative energy has been returned to the load, i.e., at a point (8) where the summation of energy (area) in the torque pulses is at its largest negative value.
- 5 To find these locations in θ corresponding to the maximum and minimum (j)'s and thus find the amount of energy needed to be stored in the flywheel, we need to numerically integrate each pulse of this function from crossover to crossover with the average line. The crossover points in Figure 11-11 have been labeled A, B, C, and D. (Program FOURBAR does this integration for you numerically, using a trapezoidal rule.)
- 6 The FOURBAR program prints the table of areas shown in Figure 11-11. The positive and negative pulses are separately integrated as described above. Reference to the plot of the torque function will indicate whether a positive or negative pulse is the first encountered in a particular case. The first pulse in this example is a positive one.
- 7 The remaining task is to accumulate these pulse areas beginning at an arbitrary crossover (in this case point A) and proceeding pulse by pulse across the cycle. Table 11-1 shows this process and the result.
- 8 Note in Table 11-1 that the minimum shaft speed occurs after the largest accumulated positive energy pulse (+200.73 in-lb) has been delivered from the driveshaft to the system. This delivery of energy slows the motor down. The maximum shaft speed occurs after the largest accumulated negative energy pulse (-60.32 in-lb) has been received back from the system by the driveshaft. This return of stored energy will speed up the motor. The total energy variation is the algebraic difference between these two extreme values, which in this example is -261.05 in-lb. This negative energy coming out of the system needs to be absorbed by the flywheel and then returned to the system *during each cycle* to smooth the variations in shaft speed.

TABLE 11-1 Integrating the Torque Function

From	$\Delta \text{ Area} = \Delta E$	Accum. Sum = E	
A to B	+200.73	+200.73	$\omega_{min} @ B$
B to C	-261.05	-60.32	$\omega_{max} @ C$
C to D	+153.88	+93.56	
D to A	-92.02	+1.54	
Total $\Delta \text{ Energy}$		$= E @ \omega_{max} - E @ \omega_{min}$ $= (-60.32) - (+200.73) = -261.05 \text{ in} \cdot \text{lb}$	

SIZING THE FLYWHEEL We now must determine how large a flywheel is needed to absorb this energy with an acceptable change in speed. The change in shaft speed during a cycle is called its *fluctuation* (Fl) and is equal to:

$$Fl = \omega_{max} - \omega_{min} \quad (11.19a)$$

We can normalize this to a dimensionless ratio by dividing it by the average shaft speed. This ratio is called the *coefficient of fluctuation* (k).

$$k = \frac{(\omega_{max} - \omega_{min})}{\omega_{avg}} \quad (11.19b)$$

This *coefficient of fluctuation* is a design parameter to be chosen by the designer. It typically is set to a value between 0.01 and 0.05, which correspond to a 1 to 5% fluctuation in shaft speed. The smaller this chosen value, the larger the flywheel will have to be. This presents a design trade-off. A larger flywheel will add more cost and weight to the system, which factors have to be weighed against the smoothness of operation desired.

We found the required change in energy E by integrating the torque curve

$$\int_{\theta @ \omega_{min}}^{\theta @ \omega_{max}} (T_L - T_{avg}) d\theta = E \quad (11.20a)$$

and can now set it equal to the right side of equation 11.18c (p. 550):

$$E = \frac{1}{2} I (\omega_{max}^2 - \omega_{min}^2) \quad (11.20b)$$

Factoring this expression:

$$E = \frac{1}{2} I (\omega_{max} + \omega_{min}) (\omega_{max} - \omega_{min}) \quad (11.20c)$$

If the torque-time function were a pure harmonic, then its average value could be expressed exactly as:

$$\omega_{avg} = \frac{(\omega_{max} + \omega_{min})}{2} \quad (11.21)$$

Our torque functions will seldom be pure harmonics, but the error introduced by using this expression as an approximation of the average is acceptably small. We can now substitute equations 11.19b and 11.21 into equation 11.20c to get an expression for the mass moment of inertia I_s of the system flywheel needed.

$$E = \frac{1}{2} I (2\omega_{avg}) (k \omega_{avg})$$

$$I_s = \frac{E}{k \omega_{avg}^2} \quad (11.22)$$

Equation 11.22 can be used to design the physical flywheel by choosing a desired coefficient of fluctuation k , and using the value of E from the numerical integration of

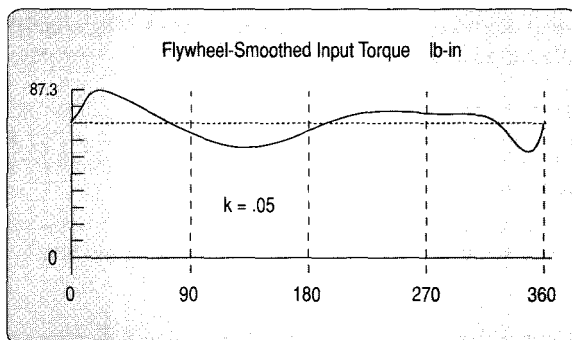


FIGURE 11-12

Input torque curve for the fourbar linkage in Figure 11-8 after smoothing with a flywheel

the torque curve (see Table 11-1, p. 552) and the average shaft ω to compute the needed system I_S . The physical flywheel's mass moment of inertia I_f is then set equal to the required system I_S . But if the moments of inertia of the other rotating elements on the same driveshaft (such as the motor) are known, the physical flywheel's required I_f can be reduced by those amounts.

The most efficient flywheel design in terms of maximizing I for minimum material used is one in which the mass is concentrated in its rim and its hub is supported on spokes, like a carriage wheel. This puts the majority of the mass at the largest radius possible and minimizes the weight for a given I . Even if a flat, solid circular disk flywheel design is chosen, either for simplicity of manufacture or to obtain a flat surface for other functions (such as an automobile clutch), the design should be done with an eye to reducing weight and thus cost. Since in general, $I = m r^2$, a thin disk of large diameter will need fewer pounds of material to obtain a given I than will a thicker disk of smaller diameter. Dense materials such as cast iron and steel are the obvious choices for a flywheel. Aluminum is seldom used. Though many metals (lead, gold, silver, platinum) are more dense than iron and steel, one can seldom get the accounting department's approval to use them in a flywheel.

Figure 11-12 shows the change in the input torque T_{12} for the linkage in Figure 11-8 after the addition of a flywheel sized to provide a coefficient of fluctuation of 0.05. The oscillation in torque about the unchanged average value is now 5%, much less than what it was without the flywheel. A much smaller horsepower motor can now be used because the flywheel is available to absorb the energy returned from the linkage during its cycle.

11.12 A LINKAGE FORCE TRANSMISSION INDEX

The transmission angle was introduced in Chapter 2 and used in subsequent chapters as an index of merit to predict the kinematic behavior of a linkage. A too-small transmission angle predicts problems with motion and force transmission in a fourbar linkage. Unfortunately, the transmission angle has limited application. It is only useful for fourbar linkages and then only when the input and output torques are applied to links that are

pivoted to ground (i.e., the crank and rocker). When external forces are applied to the coupler link, the transmission angle tells nothing about the linkage's behavior.

Holte and Chase [1] define a joint-force index (*IPI*) which is useful as an indicator of any linkage's ability to smoothly transmit energy regardless of where the loads are applied on the linkage. It is applicable to higher-order linkages as well as to the fourbar linkage. The IPI at any instantaneous position is defined as the ratio of the maximum static force in any joint of the mechanism to the applied external load. If the external load is a force, then it is:

$$JFI = \max \left| \frac{F_{ij}}{F_{ext}} \right| \quad \text{for all pairs } i, j \quad (11.23a)$$

If the external load is a torque, then it is:

$$JFI = \max \left| \frac{F_{ij}}{T_{ext}} \right| \quad \text{for all pairs } i, j \quad (11.23b)$$

where, in both cases, F_{ij} is the force in the linkage joint connecting links i and j .

The F_{ij} are calculated from a static force analysis of the linkage. Dynamic forces can be much greater than static forces if speeds are high. However, if this static force transmission index indicates a problem in the absence of any dynamic forces, then the situation will obviously be worse at speed. The largest joint force at each position is used rather than a composite or average value on the assumption that high friction in anyone joint is sufficient to hamper linkage performance regardless of the forces at other joints.

Equation 11.23a is dimensionless and so can be used to compare linkages of different design and geometry. Equation 11.23b has dimensions of reciprocal length, so caution must be exercised when comparing designs when the external load is a torque. Then the units used in any comparison must be the same, and the compared linkages should be similar in size.

Equations 11.23 apply to anyone instantaneous position of the linkage. As with the transmission angle, this index must be evaluated for all positions of the linkage over its expected range of motion and the largest value of that set found. The peak force may move from pin to pin as the linkage rotates. If the external loads vary with linkage position, they can be accounted for in the calculation.

Holte and Chase suggest that the IPI be kept below a value of about 2 for linkages whose output is a force. Larger values may be tolerable especially if the joints are designed with good bearings that are able to handle the higher loads.

There are some linkage positions in which the IPI can become infinite or indeterminate as when the linkage reaches an immovable position, defined as the input link or input joint being inactive. This is equivalent to a stationary configuration as described in earlier chapters provided that the input joint is inactive in the particular stationary configuration. These positions need to be identified and avoided in any event, independent of the determination of any index of merit. In some cases the mechanism may be immovable but still capable of supporting a load. See reference [1] for more detailed information on these special cases.

11.13 PRACTICAL CONSIDERATIONS

This chapter has presented some approaches to the computation of dynamic forces in moving machinery. The newtonian approach gives the most information and is necessary in order to obtain the forces at all pin joints so that stress analyses of the members can be done. Its application is really quite straightforward, requiring only the creation of correct free-body diagrams for each member and the application of the two simple vector equations which express Newton's second law to each free-body. Once these equations are expanded for each member in the system and placed in standard matrix form, their solution (with a computer) is a trivial task.

The real work in designing these mechanisms comes in the determination of the shapes and sizes of the members. In addition to the kinematic data, the force computation requires only the masses, CG locations, and mass moments of inertia versus those CGs for its completion. These three geometric parameters completely characterize the member for dynamic modelling purposes. Even if the link shapes and materials are completely defined at the outset of the force analysis process (as with the redesign of an existing system), it is a tedious exercise to calculate the dynamic properties of complicated shapes. Current solids modelling CAD systems make this step easy by computing these parameters automatically for any part designed within them.

If, however, you are starting from scratch with your design, the *blank-paper syndrome* will inevitably rear its ugly head. A first approximation of link shapes and selection of materials must be made in order to create the dynamic parameters needed for a "first pass" force analysis. A stress analysis of those parts, based on the calculated dynamic forces, will invariably find problems that require changes to the part shapes, thus requiring recalculation of the dynamic properties and recomputation of the dynamic forces and stresses. This process will have to be repeated in circular fashion (*iteration-see* Chapter 1, p. 8) until an acceptable design is reached. The advantages of using a computer to do these repetitive calculations is obvious and cannot be overstressed. An equation solver program such as *TKSolver* or *Mathcad* will be a useful aid in this process by reducing the amount of computer programming necessary.

Students with no design experience are often not sure how to approach this process of designing parts for dynamic applications. The following suggestions are offered to get you started. As you gain experience, you will develop your own approach.

It is often useful to create complex shapes from a combination of simple shapes, at least for first approximation dynamic models. For example, a link could be considered to be made up of a hollow cylinder at each pivot end, connected by a rectangular prism along the line of centers. It is easy to calculate the dynamic parameters for each of these simple shapes and then combine them. The steps would be as follows (repeated for each link):

- 1 Calculate the volume, mass, CG location, and mass moments of inertia with respect to the local CG of each separate part of your built-up link. In our example link these parts would be the two hollow cylinders and the rectangular prism.
- 2 Find the location of the composite CG of the assembly of the parts into the link by the method shown in Section 11.4 (p. 531) and equation 11.3 (p. 524). See also Figure 11-2 (p. 526).

- 3 Use the *parallel axis theorem* (equation 10.8, p. 497) to transfer the mass moments of inertia of each part to the common, composite *CG* for the link. Then add the individual, transferred I 's of the parts to get the total I of the link about its composite *CG*. See Section 11.6 (p. 541).

Steps 1 to 3 will create the link geometry data for each link needed for the dynamic force analysis as derived in this chapter.

- 4 Do the dynamic force analysis.
- 5 Do a dynamic stress and deflection analysis of all parts.
- 6 Redesign the parts and repeat steps 1 to 5 until a satisfactory result is achieved.

Remember that lighter (lower mass) links will have smaller inertial forces on them and thus could have lower stresses despite their smaller cross sections. Also, smaller mass moments of inertia of the links can reduce the driving torque requirements, especially at higher speeds. But be cautious about the dynamic deflections of thin, light links becoming too large. We are assuming rigid bodies in these analyses. That assumption will not be valid if the links are too flexible. Always check the deflections as well as the stresses in your designs.

11.14 REFERENCES

- 1 **Holte, J. E., and T. R. Chase.** (1994). "A Force Transmission Index for Planar Linkage Mechanisms." *Proc. of 23rd Biennial Mechanisms Conference*, Minneapolis, MN, p. 377.

11.15 PROBLEMS

- 11-1 Draw free-body diagrams of the links in the geared fivebar linkage shown in Figure 4-11 (p. 165) and write the dynamic equations to solve for all forces plus the driving torque. Assemble the symbolic equations in matrix form for solution.
- 11-2 Draw free-body diagrams of the links in the sixbar linkage shown in Figure 4-12 (p. 167) and write the dynamic equations to solve for all forces plus the driving torque. Assemble the symbolic equations in matrix form for solution.
- *†‡11-3 Table P11-1 shows kinematic and geometric data for several slider-crank linkages of the type and orientation shown in Figure P11-1. The point locations are defined as described in the text. For the row(s) in the table assigned, use the matrix method of Section 11.5 (p. 538) and program *MATRIX*, *Mathcad*, *Matlab*, *TKSolver*, or a matrix solving calculator to solve for forces and torques at the position shown. Also compute the shaking force and shaking torque. Consider the coefficient of friction μ between slider and ground to be zero. You may check your solution by opening the solution files (located in the Solutions folder on the CD-ROM) named P11-03x (where x is the row letter) into program *SLIDER*.
- *†11-4 Repeat Problem 11-3 using the method of virtual work to solve for the input torque on link 2. Additional data for corresponding rows are given in Table P11-2.
- *†11-5 Table P11-3 shows kinematic and geometric data for several pin-jointed fourbar linkages of the type and orientation shown in Figure P11-2. All have $\theta_1 = 0$. The

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

§ These problems are suited to solution using program *SLIDER* which is on the attached CD-ROM.

TABLE P11-1 Data for Problem 11-3 (See Figure P11-1 for Nomenclature)

Part 1 Lengths in inches, angles in degrees, mass in blobs, angular velocity in rad/sec

Row	link 2	link 3	offset	θ_2	ω_2	α_2	m_2	m_3	m_4
a.	4	12	0	45	10	20	0.002	0.020	0.060
b.	3	10	1	30	15	-5	0.050	0.100	0.200
c.	5	15	-1	260	20	15	0.010	0.020	0.030
d.	6	20	1	-75	-10	-10	0.006	0.150	0.050
e.	2	8	0	135	25	25	0.001	0.004	0.014
f.	10	35	2	120	5	-20	0.150	0.300	0.050
g.	7	25	-2	-45	30	-15	0.080	0.200	0.100

Part 2 Angular acceleration in rad/sec², moments of inertia in blob-in², torque in lb-in

Row	I_2	I_3	Rg_2 mag	δ_2 ang	Rg_3 mag	δ_3 ang	Fp_3 mag	δFp_3 ang	Rp_3 mag	δRp_3 ang	T_3
a.	0.10	0.2	2	0	5	0	0	0	0	0	20
b.	0.20	0.4	1	20	4	-30	10	45	4	30	-35
c.	0.05	0.1	3	-40	9	50	32	270	0	0	-65
d.	0.12	0.3	3	120	12	60	15	180	2	60	-12
e.	0.30	0.8	0.5	30	3	75	6	-60	2	75	40
f.	0.24	0.6	6	45	15	135	25	270	0	0	-75
g.	0.45	0.9	4	-45	10	225	9	120	5	45	-90

Part 3 Forces in lb, linear accelerations in inches/sec²

Row	θ_3	α_3	ag_2 mag	ag_2 ang	ag_3 mag	ag_3 ang	ag_4 mag	ag_4 ang
a.	166.40	-2.40	203.96	213.69	371.08	200.84	357.17	180
b.	177.13	34.33	225.06	231.27	589.43	200.05	711.97	180
c.	195.17	-134.76	1 200.84	37.85	2 088.04	43.43	929.12	0
d.	199.86	-29.74	301.50	230.71	511.74	74.52	23.97	180
e.	169.82	113.12	312.75	-17.29	976.79	-58.13	849.76	0
f.	169.03	3.29	192.09	23.66	302.50	-29.93	301.92	0
g.	186.78	-172.20	3 600.50	90.95	8 052.35	134.66	4 909.27	180

TABLE P11-2 Data for Problem 11-4

See also Table P11-1. Unit system is the same as in that table.

Row	ω_3	Vg_2 mag	Vg_2 ang	Vg_3 mag	Vg_3 ang	Vg_4 mag	Vg_4 ang	Vp_3 mag	Vp_3 ang
a.	-2.43	20.0	135	35.24	152.09	35.14	180	35.24	152.09
b.	-3.90	15.0	140	40.35	140.14	24.45	180	26.69	153.35
c.	1.20	60.0	310	89.61	-8.23	93.77	0	89.61	-8.23
d.	0.83	30.0	315	69.10	191.15	63.57	180	70.63	191.01
e.	4.49	12.5	255	56.02	211.93	29.01	180	61.36	204.87
f.	0.73	30.0	255	60.89	210.72	38.46	180	60.89	210.72
g.	-5.98	120.0	0	211.46	61.31	166.14	0	208.60	53.19

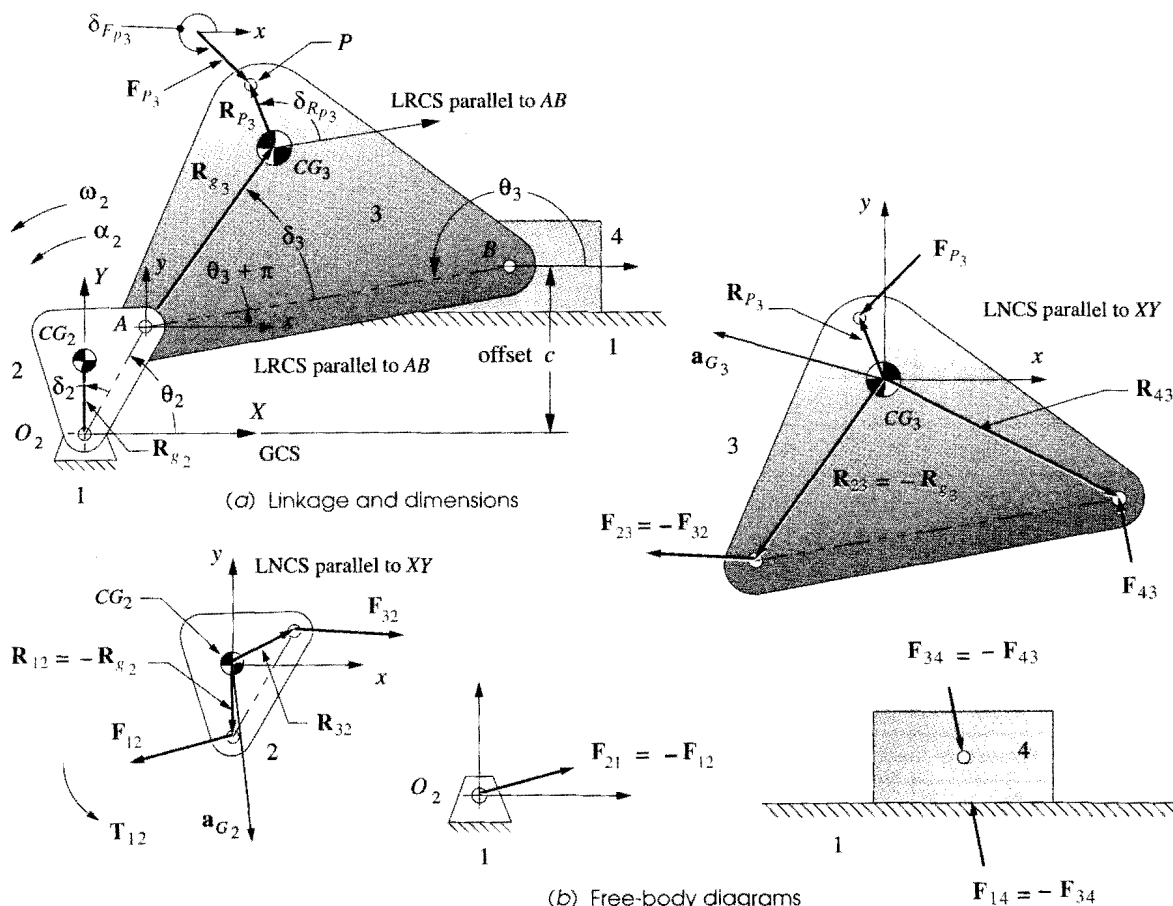


FIGURE P11-1

(b) Free-body diagrams

Linkage geometry and free body diagrams for problem 11-3

point locations are defined as described in the text. For the row(s) in the table assigned, use the matrix method of Section 11.4 (p. 531) and program MATRIX or a matrix solving calculator to solve for forces and torques at the position shown. Also compute the shaking force and shaking torque. Work in any units system you prefer. You may check your solution by opening the solution files named P11-05x (where x is the row letter) into program FOURBAR.

*†11-6 Repeat Problem 11-5 using the method of virtual work to solve for the input torque on link 2. Additional data for corresponding rows are given in Table P11-4.

*‡11-7 For the row(s) assigned in Table P11-3 (a-f), input the associated disk file to program FOURBAR, calculate the linkage parameters for crank angles from zero to 360° by 5° increments, and design a steel disk flywheel to smooth the input torque using a coefficient of fluctuation of 0.05. Minimize the flywheel weight.

‡11-8 Figure P11-3 shows a fourbar linkage and its dimensions. The steel crank and rocker have uniform cross sections 1 in wide by 0.5 in thick. The aluminum coupler is 0.75

* Answers in Appendix F.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program FOURBAR which is on the attached CD-ROM.

TABLE P11-3 Data for Problems 11-5 and 11-7

Part 1 Lengths in inches, angles in degrees, mass in blobs, angular velocity in rad/sec

Row	link 2	link 3	link 4	link 1	θ_2	θ_3	θ_4	α_2	α_3	α_4
a.	4	12	8	15	45	24.97	99.30	20	75.29	244.43
b.	3	10	12	6	30	90.15	106.60	-5	140.96	161.75
c.	5	15	14	2	260	128.70	151.03	15	78.78	53.37
d.	6	19	16	10	-75	91.82	124.44	-10	-214.84	-251.82
e.	2	8	7	9	135	34.02	122.71	25	71.54	-14.19
f.	17	35	23	4	120	348.08	19.01	-20	-101.63	-150.86
g.	7	25	10	19	-45	355.93	27.19	-15	-251.86	-661.41

Part 2 Angular acceleration in rad/sec², moments of inertia in blob-in², torque in lb-in

Row	ω_2	ω_3	ω_4	m_2	m_3	m_4	I_2	I_3	I_4	T_3	T_4
a.	20	-5.62	3.56	0.002	0.02	0.10	0.10	0.20	0.50	-15	25
b.	10	-10.31	-7.66	0.050	0.10	0.20	0.20	0.40	0.40	12	0
c.	20	16.60	14.13	0.010	0.02	0.05	0.05	0.10	0.13	-10	20
d.	20	3.90	-3.17	0.006	0.15	0.07	0.12	0.30	0.15	0	30
e.	20	1.06	5.61	0.001	0.04	0.09	0.30	0.80	0.30	25	40
f.	20	18.55	21.40	0.150	0.30	0.25	0.24	0.60	0.92	0	-25
g.	20	6.62	25.20	0.080	0.20	0.12	0.45	0.90	0.54	0	0

Part 3 Forces in lb, linear accelerations in inches/sec²

Row	R_{g_2} mag	R_{g_2} ang	R_{g_3} mag	R_{g_3} ang	R_{g_4} mag	R_{g_4} ang	a_{g_2} mag	a_{g_2} ang	a_{g_3} mag	a_{g_3} ang
a.	2	0	5	0	4	30	801.00	222.14	1 691.49	208.24
b.	1	20	4	-30	6	40	100.12	232.86	985.27	194.75
c.	3	-40	9	50	7	0	1 200.84	37.85	3 120.71	22.45
d.	3	120	12	60	6	-30	1 200.87	226.43	4 543.06	81.15
e.	0.5	30	3	75	2	-40	200.39	341.42	749.97	295.98
f.	6	45	15	135	10	25	2 403.00	347.86	12 064.20	310.22
g.	4	-45	10	225	4	45	1 601.12	202.15	2 439.12	189.26

Part 4

Row	a_{g_4} mag	a_{g_4} ang	F_{p_3} mag	δF_{p_3} ang	R_{p_3} mag	δR_{p_3} ang	F_{p_4} mag	δF_{p_4} ang	R_{p_4} mag	δR_{p_4} ang
a.	979.02	222.27	0	0	0	0	40	-30	8	0
b.	1 032.32	256.52	4	30	10	45	15	-55	12	0
c.	1 446.58	316.06	0	0	0	0	75	45	14	0
d.	1 510.34	2.15	2	45	15	180	20	270	16	0
e.	69.07	286.97	9	0	6	-60	16	60	7	0
f.	4 820.72	242.25	0	0	0	0	23	0	23	0
g.	3 667.70	298.35	12	-60	9	120	32	20	10	0

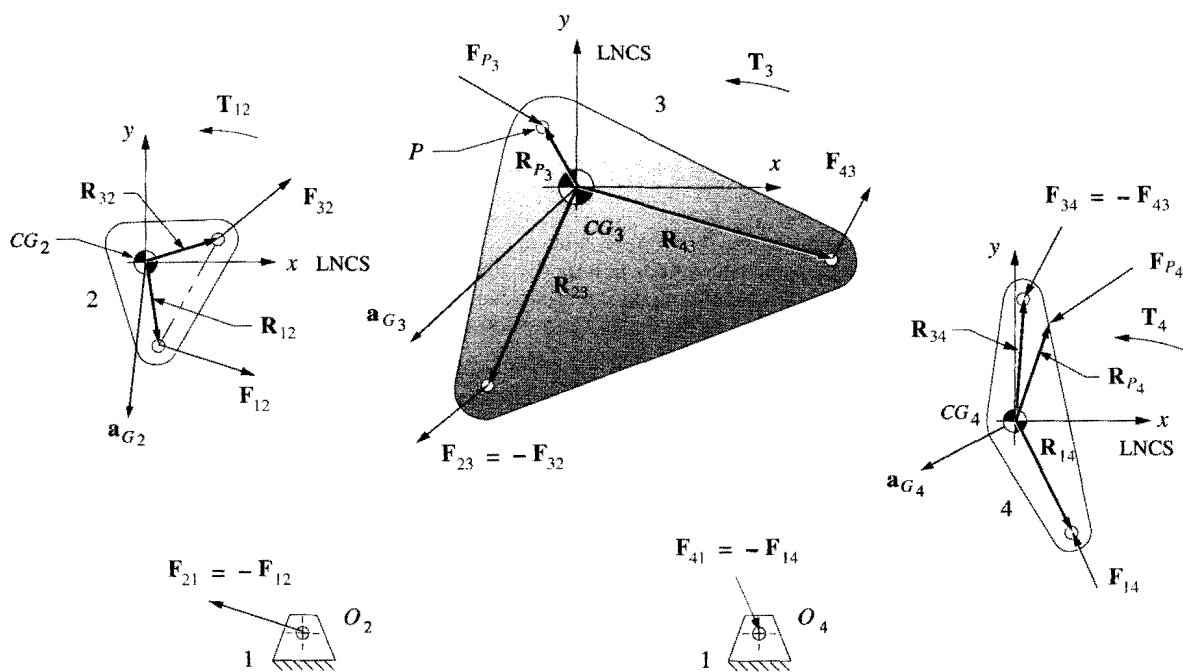
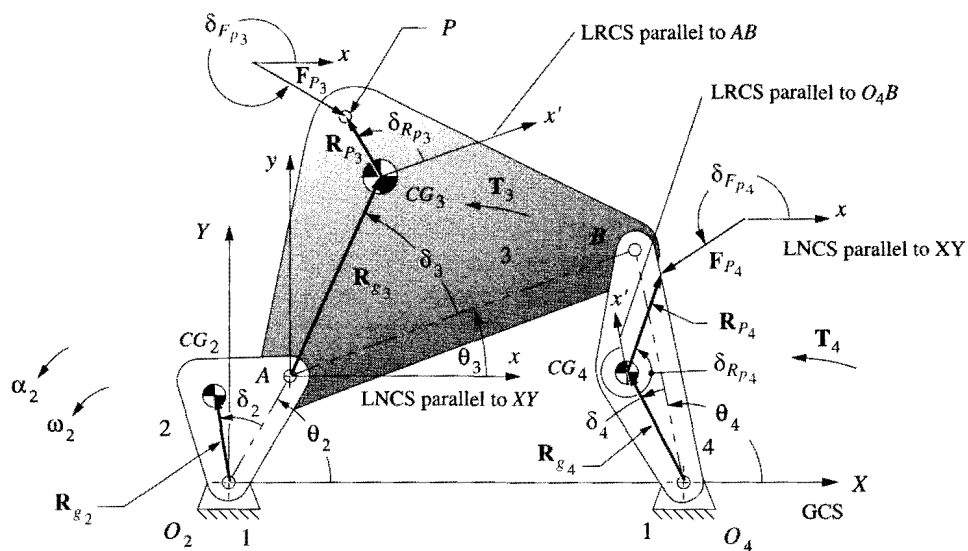
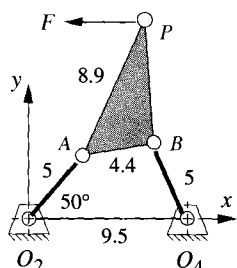


FIGURE P11-2

Linkage geometry and free-body diagrams for Problem 11-5

TABLE P11-4 Data for Problem 11-6

Row	V_{g2} mag	V_{g2} ang	V_{g3} mag	V_{g3} ang	V_{g4} mag	V_{g4} ang	V_{p3} mag	V_{p3} ang	V_{p4} mag	V_{p4} ang
a.	120.00	135.00	54.44	145.19	82.63	144.86	80.00	135.00	28.45	189.30
b.	39.55	124.96	21.46	14.74	65.15	80.91	51.05	66.14	91.89	16.60
c.	150.97	335.20	191.94	299.70	115.55	295.95	100.00	350.00	197.81	241.03
d.	103.92	45.00	94.36	353.80	138.76	13.56	113.45	12.92	50.76	34.44
e.	48.92	230.87	42.89	223.13	47.70	214.27	39.32	211.16	39.28	212.71
f.	433.24	221.29	618.05	211.39	443.42	182.08	340.00	210.00	492.15	109.01
g.	204.55	138.95	84.04	173.73	240.34	158.01	109.05	120.58	252.00	117.19



Dimensions in inches

FIGURE P11-3

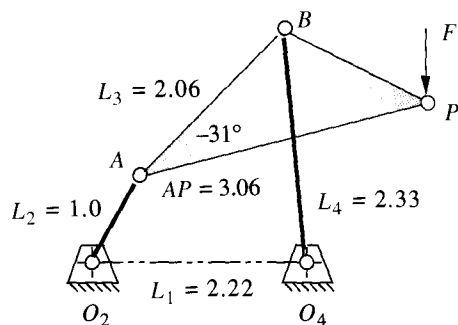
Problem 11-8

in thick. In the instantaneous position shown, the crank O_2A has $\omega = 40$ rad/sec and $\alpha = -20$ rad/sec². There is a horizontal force at P of $F = 50$ lb. Find all pin forces and the torque needed to drive the crank at this instant.

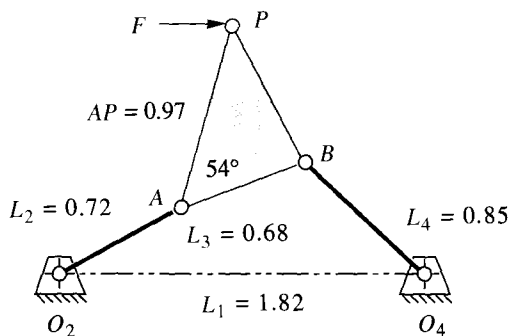
†11-9 Figure P11-4a shows a fourbar linkage and its dimensions in meters. The steel crank and rocker have uniform cross sections of 50 mm wide by 25 mm thick. The aluminum coupler is 25 mm thick. In the instantaneous position shown, the crank O_2A has $\omega = 10$ rad/sec and $\alpha = 5$ rad/sec². There is a vertical force at P of $F = 100$ N. Find all pin forces and the torque needed to drive the crank at this instant.

†11-10 Figure P11-4b shows a fourbar linkage and its dimensions in meters. The steel crank and rocker have uniform cross sections of 50 mm wide by 25 mm thick. The aluminum coupler is 25 mm thick. In the instantaneous position shown, the crank O_2A has $\omega = 15$ rad/sec and $\alpha = -10$ rad/sec². There is a horizontal force at P of $F = 200$ N. Find all pin forces and the torque needed to drive the crank at this instant.

†11-11 Figure P11-5a shows a fourbar linkage and its dimensions in meters. The steel crank, coupler, and rocker have uniform cross sections of 50 mm wide by 25 mm thick. In the instantaneous position shown, the crank O_2A has $\omega = 15$ rad/sec and $\alpha = -10$ rad/



(a)



(b)

FIGURE P11-4

Problem 11-9 to 11-10

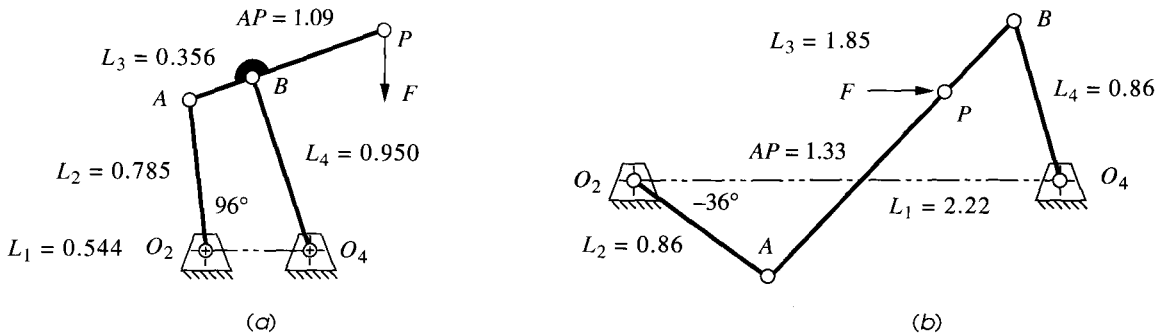


FIGURE P11-5

Problems 11-11 to 11-12

sec^2 . There is a vertical force at P of $F = 500 \text{ N}$. Find all pin forces and the torque needed to drive the crank at this instant.

*t:J11-12 Figure P11-5b shows a fourbar linkage and its dimensions in meters. The steel crank, coupler, and rocker have uniform cross sections of 50 mm diameter. In the instantaneous position shown, the crank O_2A has $\omega = -10 \text{ rad/sec}$ and $\alpha = 10 \text{ rad/sec}^2$. There is a horizontal force at P of $F = 300 \text{ N}$. Find all pin forces and the torque needed to drive the crank at this instant.

*t:J11-13 Figure P11-6 shows a water jet loom laybar drive mechanism driven by a pair of Grashof crank rocker fourbar linkages. The crank rotates at 500 rpm. The laybar is carried between the coupler-rocker joints of the two linkages at their respective instant centers h_4 . The combined weight of the reed and laybar is 29 lb. A 540-lb beat-up force from the cloth is applied to the reed as shown. The steel links have a 2×1 in uniform cross section. Find the forces on the pins for one revolution of the crank. Find the torque-time function required to drive the system.

tll-14 Figure P11-7 shows a crimping tool. Find the force $F_{\text{handneeded}}$ to generate a 2000-lb F_{crimp} . Find the pin forces. What is this linkage's joint force transmission index (IFI) in this position?

t§ 11-15 Figure P11-8 shows a walking beam conveyor mechanism that operates at slow speed (25 rpm). The boxes being pushed each weigh 50 lb. Determine the pin forces in the linkage and the torque required to drive the mechanism through one revolution. Neglect the masses of the links.

t§ 11-16 Figure P11-9 shows a surface grinder table drive that operates at 120 rpm. The crank radius is 22 mm, the coupler is 157 mm, and its offset is 40 mm. The mass of table and workpiece combined is 50 kg. Find the pin forces, slider side loads, and driving torque over one revolution.

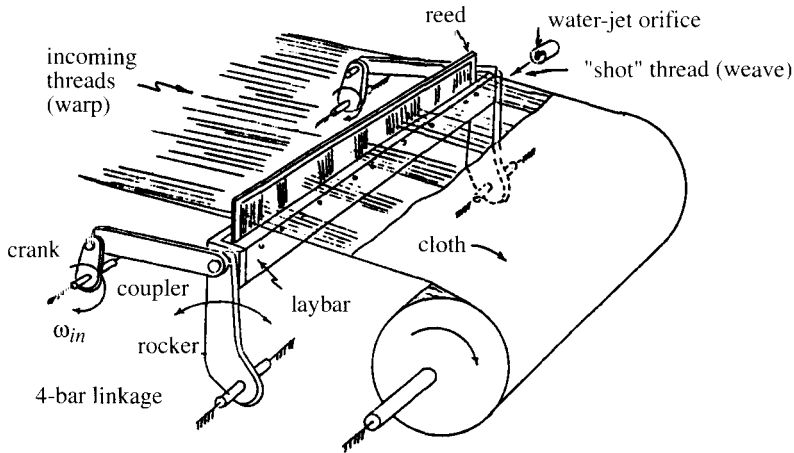
t§11-17 Figure P11-10 shows a power hacksaw that operates at 50 rpm. The crank is 75 mm, the coupler is 170 mm, and its offset is 45 mm. Find the pin forces, slider side loads, and driving torque over one revolution for a cutting force of 250 N in the forward direction and 50 N during the return stroke.

* Answers in Appendix F.

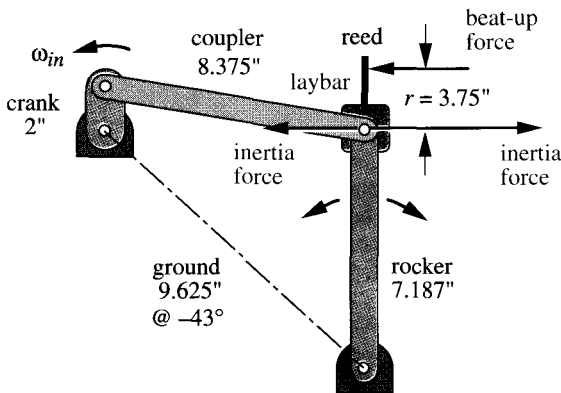
† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program *FOURBAR* which is on the attached CD-ROM.

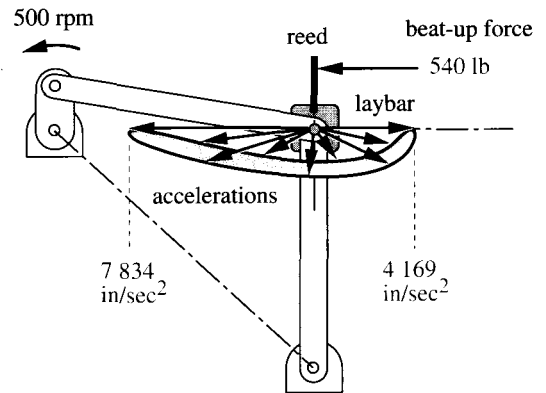
§ These problems are suited to solution using the *Working Model* program, which is on the attached CD-ROM.



(a) Warp, weave, laybar, reed, and laybar drive for a water-jet loom



(b) Linkage, laybar, reed, and dimensions



(c) Acceleration on laybar and force on reed

FIGURE P11-6

Fourbar linkage for laybar drive, showing forces and accelerations on laybar

- t§11-18 Figure P11-11 shows a paper roll off-loading station. The paper rolls have a 0.9-m OD, 0.22-m ID, are 3.23 m long, and have a density of 984 kg/m^3 . The forks that support the roll are 1.2 m long. The motion is slow so inertial loading can be neglected. Find the force required of the air cylinder to rotate the roll through 90° .
- t 11-19 Derive an expression for the relationship between flywheel mass and the dimensionless parameter radius/thickness (r/t) for a solid disk flywheel of moment of inertia I . Plot this function for an arbitrary value of I and determine the optimum r/t ratio to minimize flywheel weight for that I .

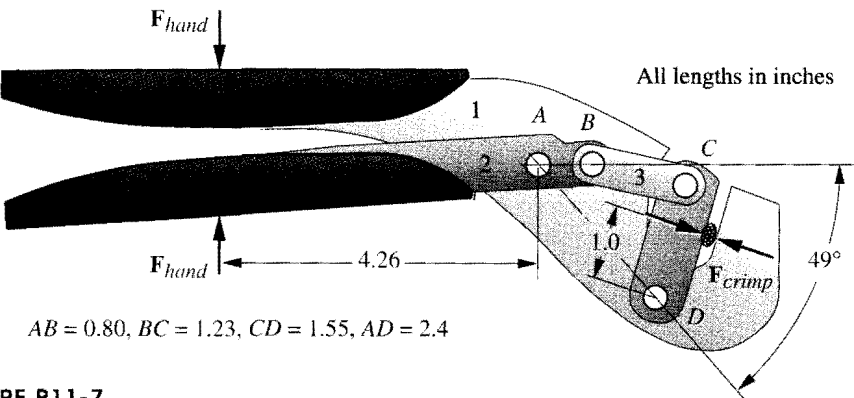


FIGURE P11-7

Problem 11-14

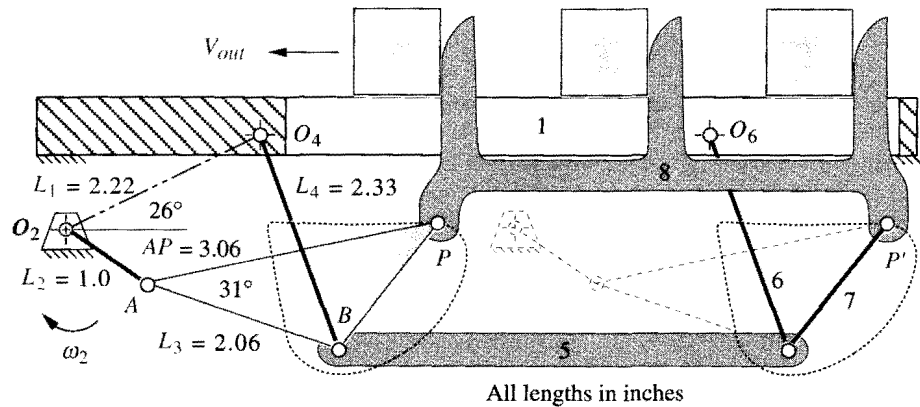


FIGURE P11-8

Problem 11-15

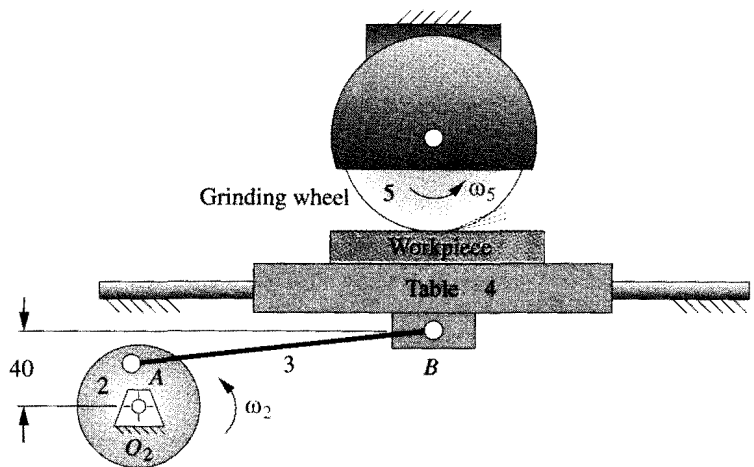
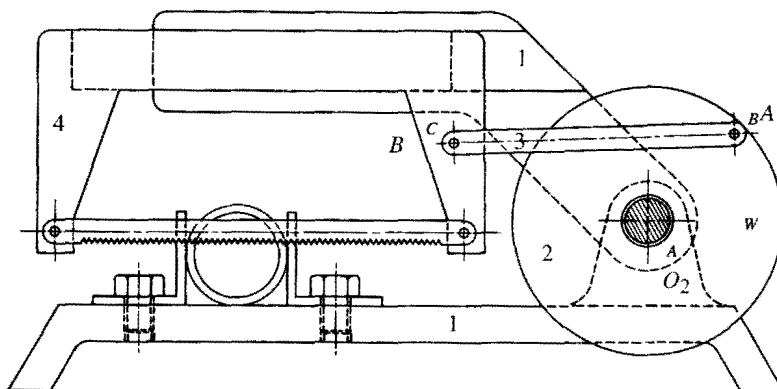


FIGURE P11-9

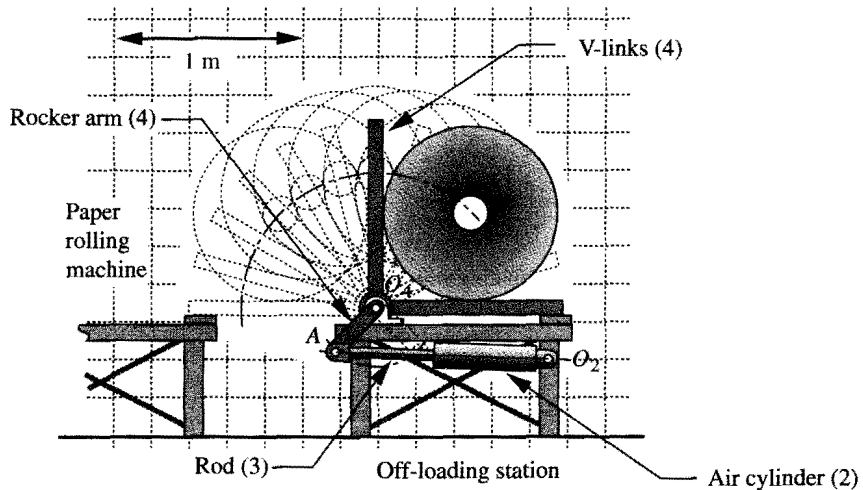
Problem 11-16

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

§ These problems are suited to solution using the *Working Model* program, which is on the attached CD-ROM.

**FIGURE P11-10**

Problem 11-17 From P. H. Hill and W. P. Rule. (1960). *Mechanisms: Analysis and Design*, with permission.

**FIGURE P11-11**

Problem 11-18

11.16 PROJECTS

The following problem statement applies to all the projects listed.

These larger-scale project statements deliberately lack detail and structure and are loosely defined. Thus, they are similar to the kind of "identification of need" or problem statement commonly encountered in engineering practice. It is left to the student to structure the problem through background research and to create a clear goal statement and set of performance specifications before attempting to design a solution. This design process is spelled out in Chapter 1 and should be followed in all of these examples. All results should be documented in a professional engineering report. See the Bibliography in Chapter 1 for references on report writing.

Some of these project problems are based on the kinematic design projects in Chapter 3. Those kinematic devices can now be designed more realistically with consideration of the dynamic forces that they generate. The strategy in most of the following project problems is to keep the dynamic pin forces and thus the shaking forces to a minimum and also keep the input torque-time curve as smooth as possible to minimize power requirements. All these problems can be solved with a pin-jointed fourbar linkage. This fact will allow you to use program FOURBAR to do the kinematic and dynamic computations on a large number and variety of designs in a short time. There are infinities of viable solutions to these problems. Iterate to find the best one! All links must be designed in detail as to their geometry (mass, moment of inertia, etc.). An equation solver such as Mathcad or TKSolver will be useful here. Determine all pin forces, shaking force, shaking torque, and input horsepower required for your designs.

PII-1 Project P3-1 stated that:

The tennis coach needs a better tennis ball server for practice. This device must fire a sequence of standard tennis balls from one side of a standard tennis court over the net such that they land and bounce within each of the three court areas defined by the court's white lines. The order and frequency of a ball's landing in anyone of the three court areas must be random. The device should operate automatically and unattended except for the refill of balls. It should be capable of firing 50 balls between reloads. The timing of ball releases should vary. For simplicity, a motor-driven pin-jointed linkage design is preferred.

This project asks you to design such a device to be mounted upon a tripod stand of 5-foot height. Design it, and the stand, for stability against tipover due to the shaking forces and shaking torques which should also be minimized in the design of your linkage. Minimize the input torque required.

PII-2 Project P3-9 stated that:

The "Save the Skeet" foundation has requested a more humane skeet launcher be designed. While they have not yet succeeded in passing legislation to prevent the wholesale slaughter of these little devils, they are concerned about the inhumane aspects of the large accelerations imparted to the skeet as it is launched into the sky for the sportsperson to shoot it down. The need is for a skeet launcher that will smoothly accelerate the clay pigeon onto its desired trajectory.

Design a skeet launcher to be mounted upon a child's "little red wagon." Control your design parameters so as to minimize the shaking forces and torques so that the wagon will remain as nearly stationary as possible during the launch of the clay pigeon.

Pl1-3 Project P3-10 stated that:

The coin-operated "kid bouncer" machines found outside supermarkets typically provide a very unimaginative rocking motion to the occupant. There is a need for a superior "bouncer" which will give more interesting motions while remaining safe for small children.

Design this device for mounting in the bed of a pickup truck. Keep the shaking forces to a minimum and the input torque-time curve as smooth as possible.

Pl1-4 Project P3-15 stated that:

NASA wants a zero-G machine for astronaut training. It must carry one person and provide a negative 1-g acceleration for as long as possible.

Design this device and its mounting hardware to the ground plane minimizing the dynamic forces and driving torque.

Pl1-5 Project P3-16 stated that:

The Amusement Machine Co. Inc. wants a portable "WHIP" ride which will give two or four passengers a thrilling but safe ride and which can be trailed behind a pickup truck from one location to another.

Design this device and its mounting hardware to the truck bed minimizing the dynamic forces and driving torque.

P 11-6 Project P3-17 stated that:

The Air Force has requested a pilot training simulator which will give potential pilots exposure to G forces similar to those they will experience in dogfight maneuvers.

Design this device and its mounting hardware to the ground plane minimizing the dynamic forces and driving torque.

Pl1-7 Project P3-18 stated that:

Cheers needs a better "mechanical bull" simulator for their "yuppie" bar in Boston. It must give a thrilling "bucking bronco" ride but be safe.

Design this device and its mounting hardware to the ground plane minimizing the dynamic forces and driving torque.

Pl1-8 Gargantuan Motors Inc. is designing a new light military transport vehicle. Their current windshield wiper linkage mechanism develops such high shaking forces when run at its highest speed that the engines are falling out! Design a superior windshield wiper mechanism to sweep the 20-lb armored wiper blade through a 90° arc while minimizing both input torque and shaking forces. The wind load on the blade, perpendicular to the windshield, is 50 lb. The coefficient of friction of the wiper blade on glass is 0.9.

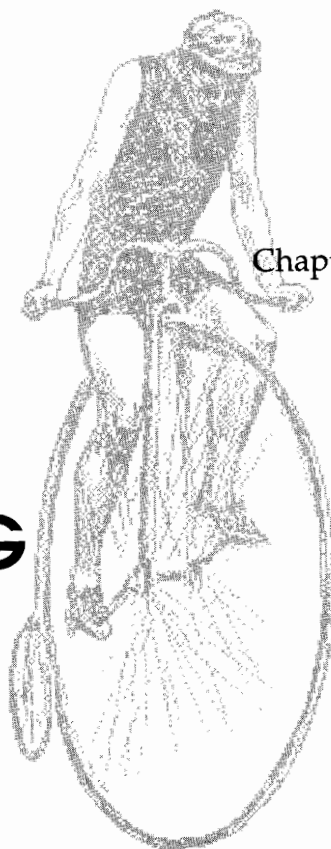
- PII-9 The Anny's latest helicopter gunship is to be fitted with the Gatling gun, which fires 50-mm-diameter, 2-cm-long spent uranium slugs at a rate of 10 rounds per second. The reaction (recoil) force may upset the chopper's stability. A mechanism is needed which can be mounted to the frame of the helicopter and which will provide a synchronous shaking force, 180° out of phase with the recoil force pulses, to counteract the recoil of the gun. Design such a linkage and minimize its torque and power drawn from the aircraft's engine. Total weight of your device should also be minimized.
- PII-10 Steel pilings are universally used as foundations for large buildings. These are often driven into the ground by hammer blows from a "pile driver." In certain soils (sandy, muddy) the piles can be "shaken" into the ground by attaching a "vibratory driver" which imparts a vertical, dynamic shaking force at or near the natural frequency of the pile-earth system. The pile can literally be made to "fall into the ground" under optimal conditions. Design a fourbar linkage-based pile shaker mechanism which, when its ground link is firmly attached to the top of a piling (supported from a crane hook), will impart a dynamic shaking force that is predominantly directed along the piling's long, vertical axis. Operating speed should be in the vicinity of the natural frequency of the pile-earth system.
- PII-11 Paint can shaker mechanisms are common in paint and hardware stores. While they do a good job of mixing the paint, they are also noisy and transmit their vibrations to the shelves and counters. A better design of the paint can shaker is possible using a balanced fourbar linkage. Design such a portable device to sit on the floor (not bolted down) and minimize the shaking forces and vibrations while still effectively mixing the paint.
- PII-12 Convertible automobiles are once again popular. While offering the pleasure of open-air motoring, they offer little protection to the occupants in a rollover accident. Permanent roll bars are ugly and detract from the open feeling of a true convertible. An automatically deployable roll bar mechanism is needed that will be out of sight until needed. In the event that sensors in the vehicle detect an imminent rollover, the mechanism should deploy within 250 ms. Design a collapsible/deployable roll bar mechanism to retrofit to the convertible of your choice.

BALANCING

*Moderation is best,
and to avoid all extremes*
PLUTARCH

Chapter

12



12.0 INTRODUCTION

Any link or member that is in pure rotation can, theoretically, be perfectly balanced to eliminate all shaking forces and shaking moments. It is accepted design practice to balance all rotating members in a machine unless shaking forces are desired (as in a vibrating shaker mechanism, for example). A rotating member can be balanced either statically or dynamically. Static balance is a subset of dynamic balance. To achieve complete balance requires that dynamic balancing be done. In some cases, static balancing can be an acceptable substitute for dynamic balancing and is generally easier to do.

Rotating parts can, and generally should, be designed to be inherently balanced by their geometry. However, the vagaries of production tolerances guarantee that there will still be some small unbalance in each part. Thus a balancing procedure will have to be applied to each part after manufacture. The amount and location of any imbalance can be measured quite accurately and compensated for by adding or removing material in the correct locations.

In this chapter we will investigate the mathematics of determining and designing a state of static and dynamic balance in rotating elements and also in mechanisms having complex motion, such as the fourbar linkage. The methods and equipment used to mea-

sure and correct imbalance in manufactured assemblies will also be discussed. It is quite convenient to use the method of d'Alembert (see Section 10.12, p. 513) when discussing rotating imbalance, applying inertia forces to the rotating elements, so we will do that.

12.1 STATIC BALANCE

Despite its name, **static balance** *does* apply to things in motion. The unbalanced forces of concern are due to the accelerations of masses in the system. The requirement for **static balance** is simply that *the sum of all forces on the moving system (including d'Alembert inertial forces) must be zero.*

$$\sum \mathbf{F} - m\mathbf{a} = 0 \quad (12.1)$$

This, of course, is simply a restatement of Newton's law as discussed in Section 10.1 (p. 491).

Another name for static balance is **single-plane balance**, which means that *the masses which are generating the inertia forces are in, or nearly in, the same plane.* It is essentially a two-dimensional problem. Some examples of common devices which meet this criterion, and thus can successfully be statically balanced, are: a single gear or pulley on a shaft, a bicycle or motorcycle tire and wheel, a thin flywheel, an airplane propeller, an individual turbine blade-wheel (but not the entire turbine). The common denominator among these devices is that they are all short in the axial direction compared to the radial direction, and thus can be considered to exist in a single plane. An automobile tire and wheel is only marginally suited to static balancing as it is reasonably thick in the axial direction compared to its diameter. Despite this fact, auto tires are sometimes statically balanced. More often they are dynamically balanced and will be discussed under that topic.

Figure 12-1a shows a link in the shape of a vee which is part of a linkage. We want to statically balance it. We can model this link dynamically as two point masses m_1 and m_2 concentrated at the local CGs of each "leg" of the link as shown in Figure 12-1b. These point masses each have a mass equal to that of the "leg" they replace and are supported on massless rods at the position (R_1 or R_2) of that leg's CG. We can solve for the required amount and location of a third "balance mass" m_b to be added to the system at some location R_b in order to satisfy equation 12.1.

Assume that the system is rotating at some constant angular velocity ω . The accelerations of the masses will then be strictly centripetal (toward the center), and the inertia forces will be centrifugal (away from the center) as shown in Figure 12-1. Since the system is rotating, the figure shows a "freeze-frame" image of it. The position at which we "stop the action" for the purpose of drawing the picture and doing the calculations is both arbitrary and irrelevant to the computation. We will set up a coordinate system with its origin at the center of rotation and resolve the inertial forces into components in that system. Writing vector equation 12.1 for this system we get:

$$-m_1 \mathbf{R}_1 \omega^2 - m_2 \mathbf{R}_2 \omega^2 - m_b \mathbf{R}_b \omega^2 = 0 \quad (12.2a)$$

Note that the only forces acting on this system are the inertia forces. For balancing, it does not matter what external forces may be acting on the system. External forces cannot be balanced by making any changes to the system's internal geometry. Note that the ω^2 terms cancel. For balancing, it also does not matter how fast the system is rotat-

ing, only that it is rotating. (The ω will determine the magnitudes of these forces, but we are going to force their sum to be zero anyway.)

Dividing

out

$$m_b \mathbf{R}_b = -m_1 \mathbf{R}_1 - m_2 \mathbf{R}_2 \quad (12.2b)$$

Breaking into x and y components:

$$\begin{aligned} m_b R_{b_x} &= -(m_1 R_{1_x} + m_2 R_{2_x}) \\ m_b R_{b_y} &= -(m_1 R_{1_y} + m_2 R_{2_y}) \end{aligned} \quad (12.2c)$$

The terms on the right sides are known. We can readily solve for the mR_x and mR_y products needed to balance the system. It will be convenient to convert the results to polar coordinates.

$$\begin{aligned} \theta_b &= \arctan \frac{m_b R_{b_y}}{m_b R_{b_x}} \\ &= \arctan \frac{-(m_1 R_{1_y} + m_2 R_{2_y})}{-(m_1 R_{1_x} + m_2 R_{2_x})} \end{aligned} \quad (12.2d)$$

$$\begin{aligned} R_b &= \sqrt{(R_{b_x})^2 + (R_{b_y})^2} \\ m_b R_b &= m_b \sqrt{(R_{b_x})^2 + (R_{b_y})^2} \\ &= \sqrt{m_b^2 (R_{b_x})^2 + m_b^2 (R_{b_y})^2} \\ &= \sqrt{m_b^2 R_{b_x}^2 + m_b^2 R_{b_y}^2} \\ &= \sqrt{(m_b R_{b_x})^2 + (m_b R_{b_y})^2} \end{aligned} \quad (12.2e)$$

The angle at which the balance mass must be placed (with respect to our arbitrarily oriented free-frame coordinate system) is θ_b , found from equation 12.2d. Note that the signs of the numerator and denominator of equation 12.2d must be individually maintained and a two-argument arctangent computed in order to obtain θ_b in the correct quadrant. Most calculators and computers will give an arctangent result only between $\pm 90^\circ$.

The $m_b R_b$ product is found from equation 12.2e. There is now an infinity of solutions available. We can either select a value for m_b and solve for the necessary radius R_b at which it should be placed, or choose a desired radius and solve for the mass that must be placed there. Packaging constraints may dictate the maximum radius possible in some cases. The balance mass is confined to the "single plane" of the unbalanced masses.

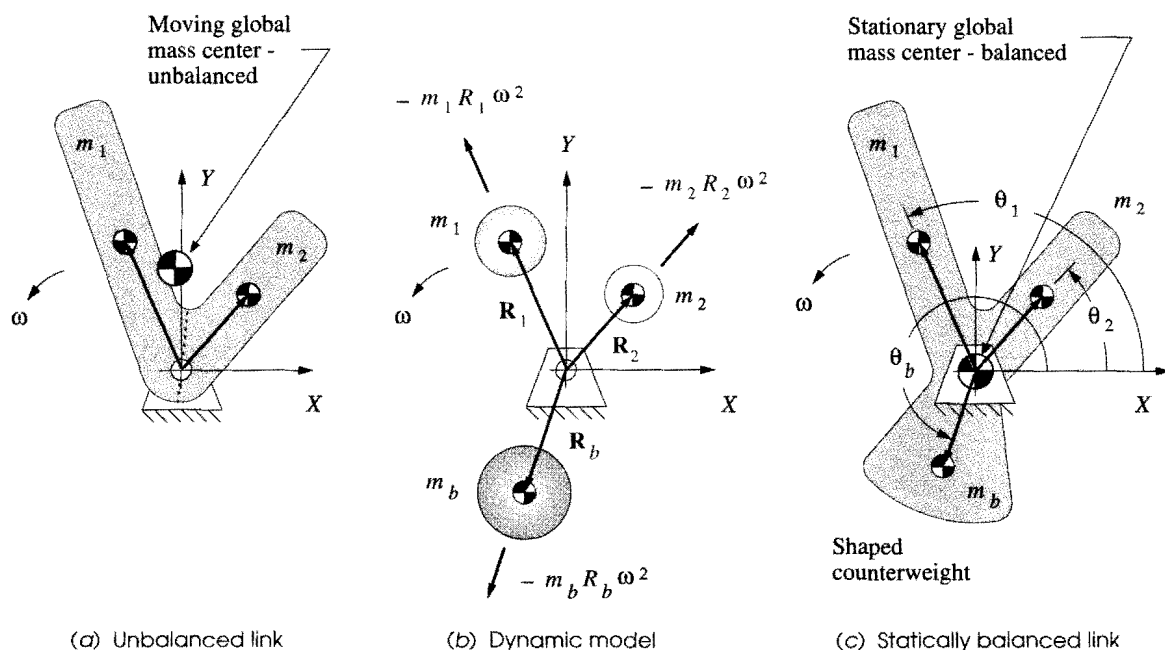


FIGURE 12-1

Static balancing a link in pure rotation

Once a combination of m_b and R_b is chosen, it remains to design the physical counterweight. The chosen radius R_b is the distance from the pivot to the *CG* of whatever shape we create for the counterweight mass. Our simple dynamic model, used to calculate the mR product, assumed a point mass and a massless rod. These ideal devices do not exist. A possible shape for this counterweight is shown in Figure 12-1c. Its mass must be m_b , distributed so as to place its *CG* at radius R_b at angle θ_b .

EXAMPLE 12-1

Static Balancing.

Given: The system shown in Figure 12-1 has the following data:

$$m_1 = 1.2 \text{ kg}$$

$$R_1 = 1.135 \text{ m @ } \angle 113.4^\circ$$

$$m_2 = 1.8 \text{ kg}$$

$$R_2 = 0.822 \text{ m @ } \angle 48.8^\circ$$

$$\omega = 40 \text{ rad/sec}$$

Find: The mass-radius product and its angular location needed to statically balance the system.

Solution:

- 1 Resolve the position vectors into xy components in the arbitrary coordinate system associated with the freeze-frame position of the linkage chosen for analysis.

$$\begin{aligned} R_1 &= 1.135 @ \angle 113.4^\circ; & R_{1x} &= -0.451, & R_{1y} &= 1.042 \\ R_2 &= 0.822 @ \angle 48.8^\circ; & R_{2x} &= +0.541, & R_{2y} &= 0.618 \end{aligned} \quad (a)$$

- 2 Solve equations 12.2c.

$$\begin{aligned} m_b R_{bx} &= -m_1 R_{1x} - m_2 R_{2x} = -(1.2)(-0.451) - (1.8)(0.541) = -0.433 \\ m_b R_{by} &= -m_1 R_{1y} - m_2 R_{2y} = -(1.2)(1.042) - (1.8)(0.618) = -2.363 \end{aligned} \quad (b)$$

- 3 Solve equations 12.2d and 12.2e.

$$\begin{aligned} \theta_b &= \arctan \frac{-2.363}{-0.433} = 259.6^\circ \\ m_b R_b &= \sqrt{(-0.433)^2 + (-2.363)^2} = 2.402 \text{ kg} \cdot \text{m} \end{aligned} \quad (c)$$

- 4 This mass-radius product of 2.402 kg-m can be obtained with a variety of shapes appended to the assembly. Figure 12-1c shows a particular shape whose CG is at a radius of $R_b = 0.806 \text{ m}$ at the required angle of 259.6° . The mass required for this counterweight design is then:

$$m_b = \frac{2.402 \text{ kg} \cdot \text{m}}{0.806 \text{ m}} = 2.980 \text{ kg} \quad (d)$$

at a chosen CG radius of:

$$R_b = 0.806 \text{ m} \quad (e)$$

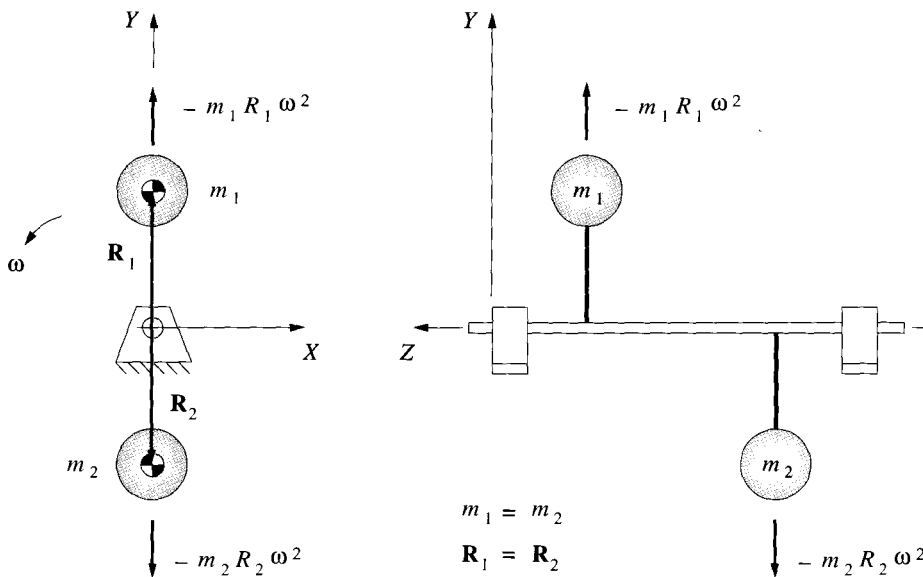
Many other shapes are possible. As long as they provide the required mass-radius product at the required angle, the system will be statically balanced. Note that the value of ω was not needed in the calculation.

12.2 DYNAMIC BALANCE

* We will use the term *moment* in this text to refer to “turning forces” whose vectors are perpendicular to an axis of rotation or “long axis” of an assembly, and the term *torque* to refer to “turning forces” whose vectors are parallel to an axis of rotation.

Dynamic balance is sometimes called **two-plane balance**. It requires that two criteria be met. The sum of the forces must be zero (static balance) plus the sum of the moments* must also be zero.

$$\begin{aligned} \sum \mathbf{F} &= 0 \\ \sum \mathbf{M} &= 0 \end{aligned} \quad (12.3)$$

**FIGURE 12-2**

Balanced forces—unbalanced moment

These moments act in planes that include the axis of rotation of the assembly such as planes XZ and YZ in Figure 12-2. The moment's vector direction, or axis, is perpendicular to the assembly's axis of rotation.

Any rotating object or assembly which is relatively long in the axial direction compared to the radial direction requires dynamic balancing for complete balance. It is possible for an object to be statically balanced but not be dynamically balanced. Consider the assembly in Figure 12-2. Two equal masses are at identical radii, 180° apart rotationally, but separated along the shaft length. A summation of $-ma$ forces due to their rotation will be always zero. However, in the side view, their inertia forces form a couple which rotates with the masses about the shaft. This rocking couple causes a moment on the ground plane, alternately lifting and dropping the left and right ends of the shaft.

Some examples of devices which require dynamic balancing are: rollers, crankshafts, camshafts, axles, clusters of multiple gears, motor rotors, turbines, propeller shafts. The common denominator among these devices is that their mass may be unevenly distributed both rotationally around their axis and also longitudinally along their axis.

To correct dynamic imbalance requires either adding or removing the right amount of mass at the proper angular locations in two *correction planes* separated by some distance along the shaft. This will create the necessary counter forces to statically balance the system and also provide a counter couple to cancel the unbalanced moment. When an automobile tire and wheel is dynamically balanced, the two correction planes are the inner and outer edges of the wheel rim. Correction weights are added at the proper locations in each of these correction planes based on a measurement of the dynamic forces generated by the unbalanced, spinning wheel.

It is always good practice to first statically balance all individual components that go into an assembly, if possible. This will reduce the amount of dynamic imbalance that must be corrected in the final assembly and also reduce the bending moment on the shaft. A common example of this situation is the aircraft turbine which consists of a number of circular turbine wheels arranged along a shaft. Since these spin at high speed, the inertia forces due to any imbalance can be very large. The individual wheels are statically balanced before being assembled to the shaft. The final assembly is then dynamically balanced.

Some devices do not lend themselves to this approach. An electric motor rotor is essentially a spool of copper wire wrapped in a complex pattern around the shaft. The mass of the wire is not uniformly distributed either rotationally or longitudinally, so it will not be balanced. It is not possible to modify the windings' local mass distribution after the fact without compromising electrical integrity. Thus the entire rotor imbalance must be countered in the two correction planes after assembly.

Consider the system of three lumped masses arranged around and along the shaft in Figure 12-3. Assume that, for some reason, they cannot be individually statically balanced within their own planes. We then create two correction planes labeled *A* and *B*. In this design example, the unbalanced masses m_1 , m_2 , m_3 and their radii R_1 , R_2 , R_3 are known along with their angular locations θ_1 , θ_2 , and θ_3 . We want to dynamically balance the system. A three-dimensional coordinate system is applied with the axis of rotation in the *Z* direction. Note that the system has again been stopped in an arbitrary freeze-frame position. Angular acceleration is assumed to be zero. The summation of forces is:

$$-m_1 \mathbf{R}_1 \omega^2 - m_2 \mathbf{R}_2 \omega^2 - m_3 \mathbf{R}_3 \omega^2 - m_A \mathbf{R}_A \omega^2 - m_B \mathbf{R}_B \omega^2 = 0 \quad (12.4a)$$

Dividing out the ω^2 and rearranging we get:

$$m_A \mathbf{R}_A + m_B \mathbf{R}_B = -m_1 \mathbf{R}_1 - m_2 \mathbf{R}_2 - m_3 \mathbf{R}_3 \quad (12.4b)$$

Breaking into *x* and *y* components:

$$\begin{aligned} m_A R_{Ax} + m_B R_{Bx} &= -m_1 R_{1x} - m_2 R_{2x} - m_3 R_{3x} \\ m_A R_{Ay} + m_B R_{By} &= -m_1 R_{1y} - m_2 R_{2y} - m_3 R_{3y} \end{aligned} \quad (12.4c)$$

Equations 12.4c have four unknowns in the form of the $m\mathbf{R}$ products at plane *A* and the $m\mathbf{R}$ products at plane *B*. To solve we need the sum of the moments equation which we can take about a point in one of the correction planes such as point *O*. The moment arm *z* distances of each force measured from plane *A* are labeled l_1 , l_2 , l_3 , l_B in the figure; thus

$$(m_B \mathbf{R}_B \omega^2) l_B = -(m_1 \mathbf{R}_1 \omega^2) l_1 - (m_2 \mathbf{R}_2 \omega^2) l_2 - (m_3 \mathbf{R}_3 \omega^2) l_3 \quad (12.4d)$$

Dividing out the ω^2 , breaking into *x* and *y* components, and rearranging:

The moment in the *XZ* plane (i.e., about the *Y* axis) is:

$$m_B R_{Bx} = \frac{-(m_1 R_{1x}) l_1 - (m_2 R_{2x}) l_2 - (m_3 R_{3x}) l_3}{l_B} \quad (12.4e)$$

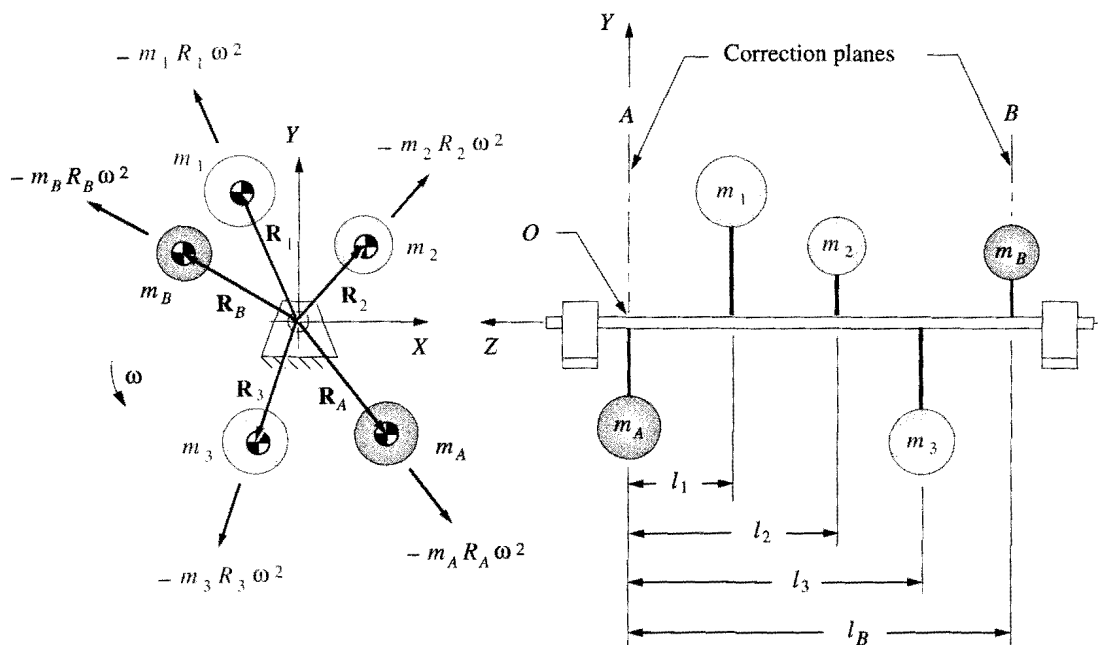


FIGURE 12-3

Two-plane dynamic balancing

The moment in the YZ plane (i.e., about the X axis) is:

$$m_B R_{B_y} = \frac{-(m_1 R_{1_y})l_1 - (m_2 R_{2_y})l_2 - (m_3 R_{3_y})l_3}{l_B} \quad (12.4f)$$

These can be solved for the mR products in x and y directions for correction plane B which can then be substituted into equation 12.4c to find the values needed in plane A . Equations 12.2d and 12.2e can then be applied to each correction plane to find the angles at which the balance masses must be placed and the mR products needed in each plane. The physical counterweights can then be designed consistent with the constraints outlined in the section on static balance. Note that the radii R_A and R_B do not have to be the same value.

EXAMPLE 12-2

Dynamic Balancing.

Given: The system shown in Figure 12-3 has the following data:

$$m_1 = 1.2 \text{ kg}$$

$$m_2 = 1.8 \text{ kg}$$

$$m_3 = 2.4 \text{ kg}$$

$$R_1 = 1.135 \text{ m @ } \angle 113.4^\circ$$

$$R_2 = 0.822 \text{ m @ } \angle 48.8^\circ$$

$$R_3 = 1.04 \text{ m @ } \angle 251.4^\circ$$

The z distances in meters from plane A are:

$$l_1 = 0.854, \quad l_2 = 1.701, \quad l_3 = 2.396, \quad l_B = 3.097$$

Find: The mass-radius products and their angular locations needed to dynamically balance the system using the correction planes A and B .

Solution:

- 1 Resolve the position vectors into xy components in the arbitrary coordinate system associated with the freeze-frame position of the linkage chosen for analysis.

$$\begin{aligned} R_1 &= 1.135 @ \angle 113.4^\circ; & R_{1x} &= -0.451, & R_{1y} &= +1.042 \\ R_2 &= 0.822 @ \angle 48.8^\circ; & R_{2x} &= +0.541, & R_{2y} &= +0.618 \\ R_3 &= 1.04 @ \angle 251.4^\circ; & R_{3x} &= -0.332, & R_{3y} &= -0.986 \end{aligned} \quad (a)$$

- 2 Solve equations 12.4e for summation of moments about point O .

$$\begin{aligned} m_B R_{Bx} &= \frac{-(m_1 R_{1x})l_1 - (m_2 R_{2x})l_2 - (m_3 R_{3x})l_3}{l_B} \\ &= \frac{-(1.2)(-0.451)(0.854) - (1.8)(0.541)(1.701) - (2.4)(-0.332)(2.396)}{3.097} = 0.230 \end{aligned} \quad (b)$$

$$\begin{aligned} m_B R_{By} &= \frac{-(m_1 R_{1y})l_1 - (m_2 R_{2y})l_2 - (m_3 R_{3y})l_3}{l_B} \\ &= \frac{-(1.2)(1.042)(0.854) - (1.8)(0.618)(1.701) - (2.4)(-0.986)(2.396)}{3.097} = 0.874 \end{aligned} \quad (c)$$

- 3 Solve equations 12.2d and 12.2e for the mass radius product in plane B .

$$\theta_B = \arctan \frac{0.874}{0.230} = 75.27^\circ \quad (d)$$

$$m_B R_B = \sqrt{(0.230)^2 + (0.874)^2} = 0.904 \text{ kg} \cdot \text{m}$$

- 4 Solve equations 12.4c for forces in x and y directions.

$$\begin{aligned} m_A R_{Ax} &= -m_1 R_{1x} - m_2 R_{2x} - m_3 R_{3x} - m_B R_{Bx} \\ m_A R_{Ay} &= -m_1 R_{1y} - m_2 R_{2y} - m_3 R_{3y} - m_B R_{By} \\ m_A R_{Ax} &= -(1.2)(-0.451) - 1.8(0.541) - (2.4)(-0.332) - 0.230 = 0.133 \\ m_A R_{Ay} &= -(1.2)(1.042) - (1.8)(0.618) - (2.4)(-0.986) - 0.874 = -0.72 \end{aligned} \quad (e)$$

- 5 Solve equations 12.2d and 12.2e for the mass radius product in plane A.

$$\theta_A = \arctan \frac{-0.872}{0.133} = -81.35^\circ$$

$$m_A R_A = \sqrt{(0.133)^2 + (-0.872)^2} = 0.882 \text{ kg} \cdot \text{m} \quad (f)$$

- 6 These mass-radius products can be obtained with a variety of shapes appended to the assembly in planes A and B. Many shapes are possible. As long as they provide the required mass-radius products at the required angles in each correction plane, the system will be dynamically balanced.

So, when the design is still on the drawing board, these simple analysis techniques can be used to determine the necessary sizes and locations of balance masses for any assembly in pure rotation for which the mass distribution is defined. This two-plane balance method can be used to dynamically balance any system in pure rotation, and all such systems should be balanced unless the purpose of the device is to create shaking forces or moments.

12.3 BALANCING LINKAGES

Many methods have been devised to balance linkages. Some achieve a complete balance of one dynamic factor, such as shaking force, at the expense of other factors such as shaking moment or driving torque. Others seek an optimum arrangement that collectively minimizes (but does not zero) shaking forces, moments, and torques for a best compromise. Lowen and Berkof, [1] and Lowen, Tepper, and Berkof [2] give comprehensive reviews of the literature on this subject up to 1983. Additional work has been done on the problem since that time, some of which is noted in the references at the end of this chapter.

Complete balance of any mechanism can be obtained by creating a second "mirror image" mechanism connected to it so as to cancel all dynamic forces and moments. Certain configurations of multicylinder internal combustion engines do this. The pistons and cranks of some cylinders cancel the inertial effects of others. We will explore these engine mechanisms in Chapter 14. However, this approach is expensive and is only justified if the added mechanism serves some second purpose such as increasing power, as in the case of additional cylinders in an engine. Adding a "dummy" mechanism whose only purpose is to cancel dynamic effects is seldom economically justifiable.

Most practical linkage balancing schemes seek to minimize or eliminate one or more of the dynamic effects (forces, moments, torques) by redistributing the mass of the existing links. This typically involves adding counterweights and/or changing the shapes of links to relocate their CGs. More elaborate schemes add geared counterweights to some links in addition to redistributing their mass. As with any design endeavor, there are trade-offs. For example, elimination of shaking forces usually increases the shaking moment and driving torque. We can only present a few approaches to this problem in the space available. The reader is directed to the literature for information on other methods.

Complete Force Balance of Linkages

The rotating links (cranks, rockers) of a linkage can be individually balanced by the rotating balance methods described in the previous section. The effects of the couplers, which are in complex motion, are more difficult to compensate for. Note that the process of statically balancing a rotating link, in effect, forces its mass center (*CG*) to be at its fixed pivot and thus stationary. In other words the condition of **static balance** can also be **defined as** one of *making the mass center stationary*. A coupler has no fixed pivot, and thus its mass center is, in general, always in motion.

Any mechanism, no matter how complex, will have, for every instantaneous position, a single, overall, *global mass center* located at some particular point. We can calculate its location knowing only the link masses and the locations of the *CGs* of the individual links at that instant. The global mass center normally will change position as the linkage moves. If we can somehow force this global mass center to be stationary, we will have a state of static balance for the overall linkage.

The Berkof-Lowen method of linearly independent vectors^[3] provides a means to calculate the magnitude and location of counterweights to be placed on the rotating links which will make the global mass center stationary for all positions of the linkage. Placement of the proper balance masses on the links will cause the dynamic forces on the fixed pivots to always be equal and opposite, i.e., a couple, thus creating static balance ($\Sigma F = 0$ but $\Sigma M \neq 0$) in the moving linkage.

This method works for any n -link planar linkage having a combination of revolute (pin) and prismatic (slider) joints, provided that there exists a path to the ground from every link which only contains revolute joints.^[4] In other words, if all possible paths from any one link to the ground contain sliding joints, then the method fails. Any linkage of n links that meets the above criterion can be balanced by the addition of $n/2$ balance weights, each on a different link.^[4] We will apply the method from reference [3] to a fourbar linkage.

Figure 12-4 shows a fourbar linkage with its overall global mass center located by the position vector $\mathbf{R}_t$. The individual *CGs* of the links are located *in the global system* by position vectors $\mathbf{R}_2$, $\mathbf{R}_3$, and $\mathbf{R}_4$ (magnitudes R_2 , R_3 , R_4), rooted at its origin, the crank pivot O_2 . The link lengths are defined by position vectors labeled $\mathbf{L}_1$, $\mathbf{L}_2$, $\mathbf{L}_3$, $\mathbf{L}_4$ (magnitudes l_1 , l_2 , l_3 , l_4), and the local position vectors which locate the *CGs within each link* are $\mathbf{B}_2$, $\mathbf{B}_3$, $\mathbf{B}_4$ (magnitudes b_2 , b_3 , b_4). The angles of the vectors $\mathbf{B}_2$, $\mathbf{B}_3$, $\mathbf{B}_4$ are ϕ_2 , ϕ_3 , ϕ_4 measured internal to the links with respect to the links' lines of centers $\mathbf{L}_2$, $\mathbf{L}_3$, $\mathbf{L}_4$. The instantaneous link angles which locate $\mathbf{L}_2$, $\mathbf{L}_3$, $\mathbf{L}_4$ in the global system are θ_2 , θ_3 , θ_4 . The total mass of the system is simply the sum of the individual link masses:

$$m_t = m_2 + m_3 + m_4 \quad (12.5a)$$

The total mass moment about the origin must be equal to the sum of the mass moments due to the individual links:

$$\Sigma M_{O_2} = m_t \mathbf{R}_t = m_2 \mathbf{R}_2 + m_3 \mathbf{R}_3 + m_4 \mathbf{R}_4 \quad (12.5b)$$

The position of the global mass center is then:

$$\mathbf{R}_t = \frac{m_2 \mathbf{R}_2 + m_3 \mathbf{R}_3 + m_4 \mathbf{R}_4}{m_t} \quad (12.5c)$$

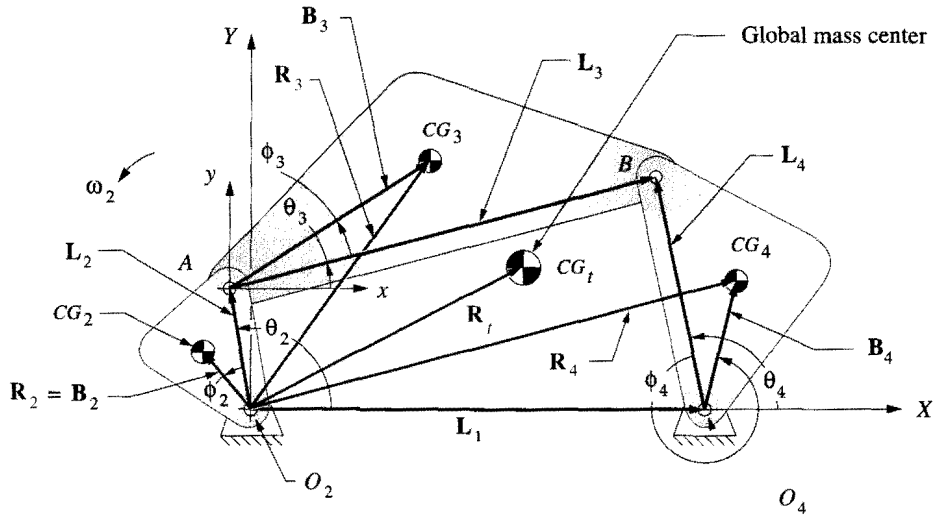


FIGURE 12-4

Static (force) balancing a fourbar linkage

and from the linkage geometry:

$$\begin{aligned} \mathbf{R}_2 &= b_2 e^{j(\theta_2 + \phi_2)} = b_2 e^{j\theta_2} e^{j\phi_2} \\ \mathbf{R}_3 &= l_2 e^{j\theta_2} + b_3 e^{j(\theta_3 + \phi_3)} = l_2 e^{j\theta_2} + b_3 e^{j\theta_3} e^{j\phi_3} \\ \mathbf{R}_4 &= l_1 e^{j\theta_1} + b_4 e^{j(\theta_4 + \phi_4)} = l_1 e^{j\theta_1} + b_4 e^{j\theta_4} e^{j\phi_4} \end{aligned} \quad (12.5d)$$

We can solve for the location of the global mass center for any link position for which we know the link angles $\theta_2, \theta_3, \theta_4$. We want to make this position vector $\mathbf{R}_t$ be a constant. The first step is to substitute equations 12.5d into 12.5b,

$$m_t \mathbf{R}_t = m_2 (b_2 e^{j\theta_2} e^{j\phi_2}) + m_3 (l_2 e^{j\theta_2} + b_3 e^{j\theta_3} e^{j\phi_3}) + m_4 (l_1 e^{j\theta_1} + b_4 e^{j\theta_4} e^{j\phi_4}) \quad (12.5e)$$

and rearrange to group the constant terms as coefficients of the time-dependent terms:

$$m_t \mathbf{R}_t = (m_4 l_1 e^{j\theta_1}) + (m_2 b_2 e^{j\phi_2} + m_3 l_2) e^{j\theta_2} + (m_3 b_3 e^{j\phi_3}) e^{j\theta_3} + (m_4 b_4 e^{j\phi_4}) e^{j\theta_4} \quad (12.5f)$$

Note that the terms in parentheses are all constant with time. The only time-dependent terms are the ones containing θ_2, θ_3 , and θ_4 .

We can also write the vector loop equation for the linkage,

$$l_2 e^{j\theta_2} + l_3 e^{j\theta_3} - l_4 e^{j\theta_4} - l_1 e^{j\theta_1} = 0 \quad (12.6a)$$

and solve it for one of the unit vectors that define a link direction, say link 3:

$$e^{j\theta_3} = \frac{(l_1 e^{j\theta_1} - l_2 e^{j\theta_2} + l_4 e^{j\theta_4})}{l_3} \quad (12.6b)$$

Substitute this into equation 12.5f to eliminate the θ_3 term and rearrange:

$$m_t \mathbf{R}_t = \left(m_2 b_2 e^{j\phi_2} + m_3 l_2 \right) e^{j\theta_2} + \frac{1}{l_3} \left(m_3 b_3 e^{j\phi_3} \right) \left(l_1 e^{j\theta_1} - l_2 e^{j\theta_2} + l_4 e^{j\theta_4} \right) \\ + \left(m_4 b_4 e^{j\phi_4} \right) e^{j\theta_4} + \left(m_4 l_1 e^{j\theta_1} \right) \quad (12.7a)$$

and collect terms:

$$m_t \mathbf{R}_t = \left(m_2 b_2 e^{j\phi_2} + m_3 l_2 - m_3 b_3 \frac{l_2}{l_3} e^{j\phi_3} \right) e^{j\theta_2} + \left(m_4 b_4 e^{j\phi_4} + m_3 b_3 \frac{l_4}{l_3} e^{j\phi_3} \right) e^{j\theta_4} \\ + m_4 l_1 e^{j\theta_1} + m_3 b_3 \frac{l_1}{l_3} e^{j\phi_3} e^{j\theta_1} \quad (12.7b)$$

This expression gives us the tool to force $\mathbf{R}_t$ to be a constant and make the linkage mass center stationary. For that to be so, the terms in parentheses which multiply the only two time-dependent variables, θ_2 and θ_4 , must be forced to be zero. (The fixed link angle θ_1 is a constant.) Thus the requirement for linkage force balance is:

$$\left(m_2 b_2 e^{j\phi_2} + m_3 l_2 - m_3 b_3 \frac{l_2}{l_3} e^{j\phi_3} \right) = 0 \quad (12.8a) \\ \left(m_4 b_4 e^{j\phi_4} + m_3 b_3 \frac{l_4}{l_3} e^{j\phi_3} \right) = 0$$

Rearrange to isolate one link's terms (say link 3) on one side of each of these equations:

$$m_2 b_2 e^{j\phi_2} = m_3 \left(b_3 \frac{l_2}{l_3} e^{j\phi_3} - l_2 \right) \quad (12.8b) \\ m_4 b_4 e^{j\phi_4} = -m_3 b_3 \frac{l_4}{l_3} e^{j\phi_3}$$

We now have two equations involving three links. The parameters for anyone link can be assumed and the other two solved for. A linkage is typically first designed to satisfy the required motion and packaging constraints before this force balancing procedure is attempted. In that event, the link geometry and masses are already defined, at least in a preliminary way. A useful strategy is to leave the link 3 mass and CG location as originally designed and calculate the necessary masses and CG locations of links 2 and 4 to satisfy these conditions for balanced forces. Links 2 and 4 are in pure rotation, so it is straightforward to add counterweights to them in order to move their CGs to the necessary locations. With this approach, the right sides of equations 12.8b are reducible to numbers for a designed linkage. We want to solve for the mass radius products $m_2 b_2$ and $m_4 b_4$ and also for the angular locations of the CGs within the links. Note that the angles ϕ_2 and ϕ_4 in equation 12.8 are measured with respect to the lines of centers of their respective links.

Equations 12.8b are vector equations. Substitute the Euler identity (equation 4.4a, p. 155) to separate into real and imaginary components, and solve for the x and y components of the mass-radius products.

$$(m_2 b_2)_x = m_3 \left(b_3 \frac{l_2}{l_3} \cos \phi_3 - l_2 \right) \quad (12.8c)$$

$$(m_2 b_2)_y = m_3 \left(b_3 \frac{l_2}{l_3} \sin \phi_3 \right)$$

$$(m_4 b_4)_x = -m_3 b_3 \frac{l_4}{l_3} \cos \phi_3 \quad (12.8d)$$

$$(m_4 b_4)_y = -m_3 b_3 \frac{l_4}{l_3} \sin \phi_3$$

These components of the mR product needed to force balance the linkage represent the entire amount needed. If links 2 and 4 are already designed with some individual unbalance (the CG not at pivot), then the existing mR product of the unbalanced link must be subtracted from that found in equations 12.8c and 12.8d in order to determine the size and location of additional counterweights to be added to those links. As we did with the balance of rotating links, any combination of mass and radius that gives the desired product is acceptable. Use equations 12.2d and 12.2e to convert the cartesian mR products in equations 12.8c and 12.8d to polar coordinates in order to find the magnitude and angle of the counterweight's mR vector. Note that the angle of the mR vector for each link will be referenced to that link's line of centers. Design the shape of the physical counterweights to be put on the links as discussed in Section 12.1 (p. 571).

12.4 EFFECT OF BALANCING ON SHAKING AND PIN FORCES

Figure 12-5 shows a fourbar linkage* to which balance masses have been added in accord with equations 12.8. Note the counterweights placed on links 2 and 4 at the calculated locations for complete force balance. Figure 12-6a shows a polar plot of the shaking forces of this linkage without the balance masses. The maximum is 462 lb at 15°. Figure 12-6b shows the shaking forces after the balance masses are added. The shaking forces are reduced to essentially zero. The small residual forces seen in Figure 12-6b are due to computational round-off errors—the method gives theoretically exact results.

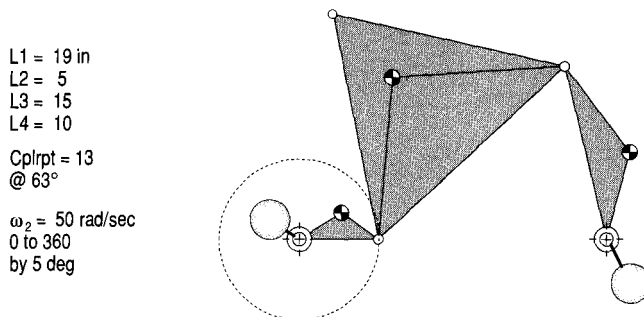
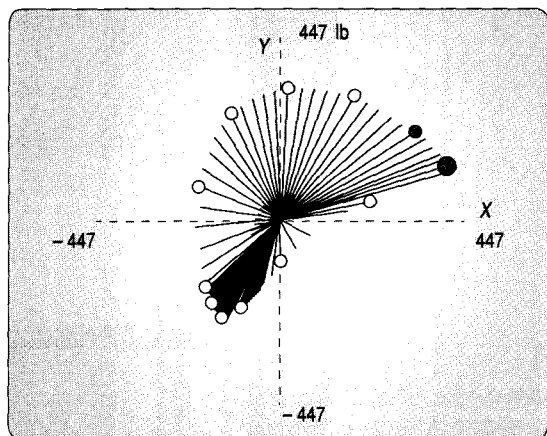


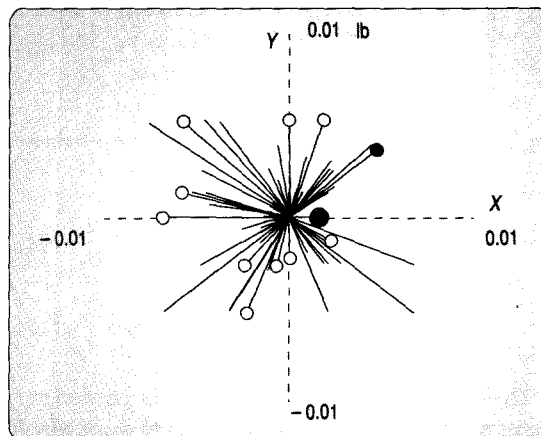
FIGURE 12-5

A balanced fourbar linkage showing balance masses applied to links 2 and 4

* Open the disk file F12-05.4br in program FOURBAR to see more detail on this linkage and its balancing.



(a) Shaking force with linkage unbalanced

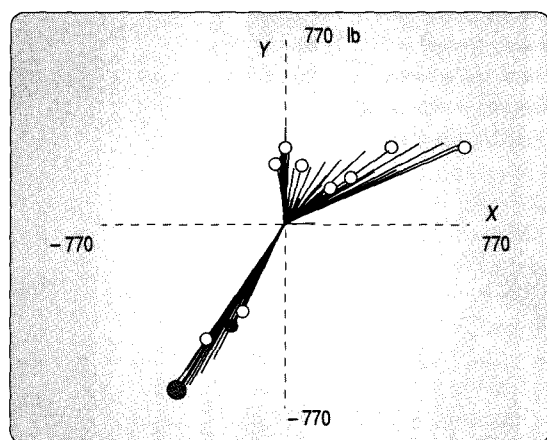


(b) Shaking force with linkage balanced

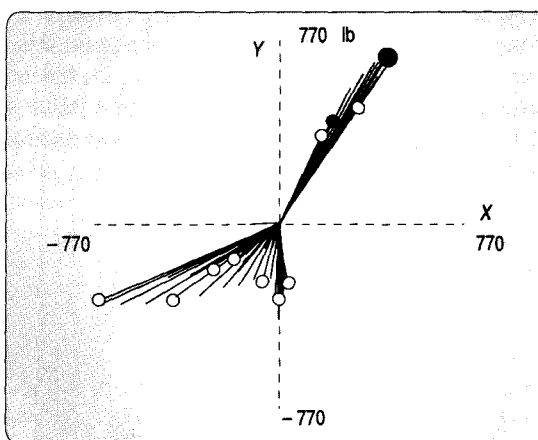
FIGURE 12-6

Polar plot of unbalanced shaking forces on ground plane of the fourbar linkage of Figure 12-5

The pin forces at the crank and rocker pivots have not disappeared as a result of adding the balance masses, however. Figures 12-7a and 12-7b, respectively, show the forces on crank and rocker pivots after balancing. These forces are now equal and opposite. After balancing, the pattern of forces at pivot O_2 is the mirror image of the pattern at pivot O_4 . The **net shaking force** is the vector sum of these two sets of forces for each time step (Section 11.8, p. 544). The equal and opposite pairs of forces acting at the



(a) Crank pivot force F21



(b) Rocker pivot force F41

FIGURE 12-7

Polar plots of forces F21 and F41 acting on the ground plane of the force-balanced fourbar linkage of Figure 12-5